

Multireference Hedin Equations

We develop a formal and practical framework for multireference Hedin equations built around a CASSCF wave function and the Dyall Hamiltonian. The first objective is to formulate an exact Dyall-reference Hedin hierarchy and to show that the multireference GW approximation of Wang, Fang, and Li can be understood as a channel-selective vertex and self-energy truncation of this hierarchy. The second objective is to go beyond that approximation at minimal additional cost by restoring the mixed Hall self-energy blocks, screening them with a static kernel consistent with the parent MR- GW approximation, and embedding the resulting couplings in a common Hermitian representation. The latter construction is causal, has a positive-semidefinite spectral function, preserves the zeroth spectral moment, and restores the exact first-order contribution to the first spectral moment. We also give a matrix-free computational formulation whose leading dense orbital scaling is unchanged relative to the parent spectral MR- GW implementation.

I. MULTIREFERENCE GREEN'S FUNCTION METHODS

Assuming the exact self-energy, the one-particle Green's function can be obtained from the Dyson equation

$$\mathbf{G}(\omega) = \mathbf{G}_0(\omega) + \mathbf{G}_0(\omega)\mathbf{\Sigma}(\omega)\mathbf{G}(\omega), \quad (1)$$

where $\mathbf{G}_0$ is the non-interacting Green's function and $\mathbf{\Sigma}$ is the exact self-energy. Now we can partition the self-energy into two contributions $\mathbf{\Sigma} = \mathbf{\Sigma}^{\text{init}} + \mathbf{\Sigma}^{\text{corr}}$, where $\mathbf{\Sigma}^{\text{init}}$ captures the contributions due to some initial treatment of correlation effects and $\mathbf{\Sigma}^{\text{corr}}$ contains all the effects that are not contained within $\mathbf{\Sigma}^{\text{init}}$.

The Dyson equation becomes

$$\mathbf{G}(\omega) = \mathbf{G}_0(\omega) + \mathbf{G}_0(\omega)\left(\mathbf{\Sigma}^{\text{init}}(\omega) + \mathbf{\Sigma}^{\text{corr}}(\omega)\right)\mathbf{G}(\omega). \quad (2)$$

After inversion, one finds

$$\mathbf{G}(\omega) = \frac{1}{\mathbf{G}_0^{-1}(\omega) - \mathbf{\Sigma}^{\text{init}}(\omega) - \mathbf{\Sigma}^{\text{corr}}(\omega)}, \quad (3)$$

where the new initial Green's function is defined as

$$\mathbf{G}_{\text{init}}(\omega) = \frac{1}{\mathbf{G}_0^{-1}(\omega) - \mathbf{\Sigma}^{\text{init}}(\omega)}. \quad (4)$$

The full Green's function can be expressed in terms of the initial Green's function and the correlation self-energy as

$$\begin{aligned} \mathbf{G}(\omega) &= \frac{1}{\mathbf{G}_{\text{init}}^{-1}(\omega) - \mathbf{\Sigma}^{\text{corr}}(\omega)} \\ &= \mathbf{G}_{\text{init}}(\omega) + \mathbf{G}_{\text{init}}(\omega)\mathbf{\Sigma}^{\text{corr}}(\omega)\mathbf{G}(\omega). \end{aligned} \quad (5)$$

Next we assume that the initial $\mathbf{\Sigma}^{\text{init}}$ is obtained from an active space calculation, e.g., for a set of orbitals $\{\phi_p\}$ defining the active space. The total orbital space can be partitioned into active and inactive orbitals, so that the ground-state wavefunction is

$$|\Psi_0\rangle = |\Phi_i\rangle \otimes |\Psi_a\rangle, \quad (6)$$

where i and a denote the inactive and active orbitals, respectively.

Within the chosen partitioning, the initial Green's function $\mathbf{G}_{\text{init}}$ can also be directly determined without resorting to the Dyson equation, for example, by using the spectral representation of the active space Hamiltonian in the $N \pm 1$ particle

sectors in combination with the coupling between the inactive orbitals to the active space. $\mathbf{G}_{\text{init}}$ reads

$$\mathbf{G}_{\text{init}}(\omega) = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & \mathbf{M} \end{pmatrix} \begin{pmatrix} \omega\mathbf{1} - \mathbf{F}_{i,i}^{\text{HF}} & \mathbf{H}_{i,a} \\ \mathbf{H}_{i,a}^\dagger & \omega\mathbf{1} - \mathbf{E}^{N\pm 1} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{1} & 0 \\ 0 & \mathbf{M}^\dagger \end{pmatrix}, \quad (7)$$

with $\mathbf{M}$ denoting the Dyson amplitudes within the active space, e.g.,

$$\begin{aligned} M_{p,\mu}^{N+1} &= \langle p | R_\mu^{N+1} \rangle = \langle \Psi_0 | [a_p, \hat{R}_\mu^{N+1}] | \Psi_0 \rangle = \langle \Psi_0 | a_p | \Psi_\mu^{N+1} \rangle, \\ M_{p,\mu}^{N-1} &= \langle p | R_\mu^{N-1} \rangle = \langle \Psi_0 | [a_p^\dagger, \hat{R}_\mu^{N-1}] | \Psi_0 \rangle = \langle \Psi_0 | a_p^\dagger | \Psi_\mu^{N-1} \rangle, \end{aligned}$$

where $(a|b)$ denotes the binary product

$$(a|b) = \langle \Psi_0 | [a^\dagger, b] | \Psi_0 \rangle. \quad (8)$$

The excitation operators $\hat{R}_\mu^{N\pm 1}$ act only within the active space, and satisfies the 'killer condition', i.e.,

$$\langle \Psi_0 | \hat{R}_\mu^{N+1} = 0, \quad \langle \Psi_0 | \hat{R}_\mu^{N-1} = 0. \quad (9)$$

Finally, $\mathbf{E}^{N\pm 1}$ denotes a diagonal matrix with the excitation energies of the active space in the $N \pm 1$ particle sectors on the diagonal.

Note that there are no Dyson amplitudes that couple the active and inactive spaces directly. However, there are Hamiltonian elements $\mathbf{H}_{i,a}$ that couple the inactive and active spaces. Analytic expressions can be obtained by defining the excitation manifolds of the inactive orbitals as

$$\{\hat{C}^\dagger\} = \{a_i, a_a\}. \quad (10)$$

The matrix elements $\mathbf{H}_{i,a}$ read

$$\mathbf{H}_{i,a} = \langle \Psi_0 | [\hat{C}, [\hat{H}, \hat{R}_\mu^{N\pm 1}]] | \Psi_0 \rangle. \quad (11)$$

One has to be a little bit careful. There might be some problem with the sign convention Since the Hamiltonian contains only one- and two-body terms, only a limited number of excitation manifolds in the excitation operators $\hat{R}_\mu^{N\pm 1}$ need to be explicitly considered.

With this, one can rewrite Eq. (7) as

$$\mathbf{G}_{\text{init}}(\omega) = \begin{pmatrix} \mathbf{G}_i(\omega) & \mathbf{G}_i(\omega)\mathbf{\Sigma}_{i,a}(\omega) \\ \mathbf{\Sigma}_{a,i}(\omega)\mathbf{G}_i(\omega) & \mathbf{G}_a + \mathbf{\Sigma}_{a,i}(\omega)\mathbf{G}_i(\omega)\mathbf{\Sigma}_{i,a}(\omega) \end{pmatrix}, \quad (12)$$

where

$$\Sigma_{i,a}(\omega) = -\mathbf{H}_{i,a} \left(\omega - \mathbf{E}^{N\pm 1} \right)^{-1} \mathbf{M},$$

and

$$\mathbf{G}_i(\omega) = \left[\mathbf{G}_{i,0}^{-1}(\omega) - \Sigma_{i,i}(\omega) \right]^{-1} = \mathbf{G}_{i,0}(\omega) + \tilde{\Sigma}_{i,i}(\omega), \quad (13)$$

with

$$\Sigma_{i,i}(\omega) = \mathbf{H}_{i,a} \left(\omega - \mathbf{E}^{N\pm 1} \right)^{-1} \mathbf{H}_{a,i}. \quad (14)$$

Eq. (12) can be compactly written as

$$\mathbf{G}_{\text{init}} = (\mathbf{1} - \Sigma_{a,i}(\omega))^{-1} \left(\mathbf{G}_{i,0}(\omega) + \tilde{\Sigma}_{i,i}(\omega) \right) (\mathbf{1} - \Sigma_{i,a}(\omega))^{-1}, \quad (15)$$

with

$$(\mathbf{1} - \Sigma_{a,i}(\omega))^{-1} = \begin{pmatrix} \mathbf{1} & 0 \\ -\Sigma_{a,i}(\omega) & \mathbf{1} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{1} & 0 \\ \Sigma_{a,i}(\omega) & \mathbf{1} \end{pmatrix}. \quad (16)$$

Eq. (15) has the same structure as the initial Dyson equation in the MR-*GW* paper. However, in this derivation the meaning of the different terms becomes clearer. In particular, we have contributions reflecting the fact that an initial correlated state $|\Psi_0\rangle$ is used. Furthermore, all remaining contributions in Σ^{corr} describe two additional correlation effects missing in G_{init} :

- Effects within the inactive space itself.
- Couplings between the inactive and active spaces that go beyond the operator manifold defined in Eq. (10).

(PFL: These notes have been automatically generated by ChatGPT after a very lengthy conversation. They need to be carefully understood, checked, and polished.)

II. INTRODUCTION AND OBJECTIVES

The aim of these notes is to develop a multireference extension of Hedin’s framework in which the zeroth-order Hamiltonian is the Dyall Hamiltonian associated with a CASSCF wave function. The central difficulty is that the Dyall reference is interacting and therefore non-Gaussian: its exact generating functional contains connected cumulants C_n^D of all orders. Consequently, the standard five-equation Hedin construction cannot simply be reused without modification.

Two complementary objectives are pursued.

a. Objective I: formal embedding of existing MR-GW. We first construct an exact Dyall-reference Hedin hierarchy by combining the exact correlated Dyall generating functional with an enlarged source-space representation inspired by Brouder’s treatment of general correlated references¹. We then verify its first-order expansion against the four generalized self-energy blocks derived by Wang, Fang, and Li², and show that their practical MR-GW scheme can be understood as a channel-selective truncation of the resulting hierarchy.

b. Objective II: a minimal nonzero-block extension. We next restore the lowest-cost information discarded by practical MR-GW, namely the mixed Hall blocks Σ^{12} and Σ^{21} . We construct a static screened kernel consistent with the parent MR-GW self-energy under a frozen- W approximation, and embed the resulting mixed couplings together with the dynamic MR-GW self-energy in a common Hermitian pole-space representation. This defines a causal positive-semidefinite completion that preserves the zeroth spectral moment and is exact through first residual order for the first moment.

Throughout the notes, the Hamiltonian partition

$$\hat{H} = \hat{H}_D + \hat{V}_R \quad (17)$$

is used without exception. The active-space interaction is treated nonperturbatively in $\hat{H}_D$, while only the residual interaction $\hat{V}_R$ is screened and expanded.

A. Status of statements

To keep the logical status of the various equations explicit, we use the following terminology.

Exact identity. denotes an identity within the exact Dyall-reference or enlarged-space formalism.

Established result. denotes a result already present in the literature, most notably the generalized Dyson construction of Hall and Brouder and the first-order self-energy/MR-GW formulas of Wang, Fang, and Li.

Proposed approximation. denotes an approximation introduced in these notes. In particular, the static mixed kernel and its finite Hermitian/PSD resummation belong to this category. Some algebraic properties of a proposed construction, such as

its Hermiticity or its spectral-moment consequences, may of course be exact once that construction has been defined.

B. Organization

Sections III and IV correspond to the two objectives above. A consolidated set of limiting cases is collected in Sec. V, followed by conclusions and the main directions for further work.

III. OBJECTIVE I: DYALL-REFERENCE HEDIN THEORY AND THE EXISTING MR-GW SCHEME

A. Exact Dyall-reference Hedin hierarchy

Exact identity. The equations in this subsection are identities of the enlarged Dyall-reference theory once the auxiliary cumulant interaction is defined by the source-transfer identity introduced below. Brouder’s construction for a general correlated state and its connection to Hall’s generalized Dyson equation are established in the literature; applying the same algebraic source transfer to the exact generating functional of the interacting Dyall reference is the extension developed here.

Our starting point is a partition of the physical Hamiltonian into a Dyall zeroth-order Hamiltonian and a residual interaction,

$$\hat{H} = \hat{H}_D + \hat{V}_R \quad (18)$$

The Dyall zeroth-order Hamiltonian is associated with a chosen CASSCF reference and contains the full interaction within the active orbital space. In contrast with the usual Hartree–Fock reference used in conventional many-body perturbation theory, $\hat{H}_D$ is therefore itself an interacting Hamiltonian. We use “Dyall Hamiltonian” in this sense throughout, rather than identifying it with the conventional CASPT2 zeroth-order Hamiltonian.

This distinction is essential. The exact generating functional associated with $\hat{H}_D$ is non-Gaussian and contains connected n -particle Green’s functions of arbitrary order. The purpose of this section is to show how this non-Gaussian reference can nevertheless be embedded into a Hedin-like framework by introducing an enlarged source space following the general strategy introduced by Brouder¹. The physical 11 sector of the resulting theory reproduces Hall’s generalized Dyson equation and therefore provides the natural formal setting for the multireference Green’s-function theory of Wang, Fang, and Li².

1. Non-Gaussian Dyall reference

Let

$$G_D(1, 2) = -i \langle T_C \hat{\psi}_D(1) \hat{\psi}_D^\dagger(2) | T_C \hat{\psi}_D(1) \hat{\psi}_D^\dagger(2) \rangle \quad (19)$$

denote the exact one-particle Green's function of the Dyall Hamiltonian, where C denotes the chosen time contour.

Introducing Grassmann sources $\bar{\eta}$ and η , the logarithm of the exact Dyall generating functional can be decomposed as

$$\ln Z_D[\bar{\eta}, \eta] = -i\bar{\eta}G_D\eta + \rho_D[\bar{\eta}, \eta] \quad (20)$$

The first term contains the exact one-particle propagator, while ρ_D contains all connected reference correlations beyond the one-particle level.

Particle-number conservation implies the expansion

$$\rho_D = \rho_D^{(2)} + \rho_D^{(3)} + \dots \quad (21)$$

where $\rho_D^{(n)}$ is homogeneous of degree n in $\bar{\eta}$ and degree n in η . Its coefficients are the connected Dyall n -particle Green's functions,

$$C_n^D \equiv G_{D,c}^{(n)} \quad (22)$$

For a Gaussian reference,

$$C_{n \geq 2}^D = 0 \quad (23)$$

whereas for a genuine CASSCF/Dyall reference one generally has

$$C_2^D C_3^D \dots \neq 0 \quad (24)$$

This is the precise field-theoretical meaning of the non-Gaussian character of the multireference zeroth-order problem.

2. Enlarged Gaussian representation

Brouder showed that the cumulants of a general correlated state can be transferred from the reference generating functional to auxiliary interaction vertices by introducing an enlarged source space¹. In Brouder's original formulation the underlying forward/backward contour (or source) structure is explicit; below we suppress these contour indices and retain only the compact 2×2 auxiliary-space structure. We use the same source-doubling idea algebraically, but apply it to the exact generating functional of the interacting Dyall reference rather than to a quadratic reference Hamiltonian. The following representation is therefore exact once the auxiliary cumulant interaction is defined by Eq. (29).

Introduce a second pair of Grassmann sources,

$$\bar{\xi} \quad \xi \quad (25)$$

and define

$$\bar{\mathcal{J}} = \begin{pmatrix} \bar{\eta} & \bar{\xi} \end{pmatrix} \quad \mathcal{J} = \begin{pmatrix} \eta \\ \xi \end{pmatrix} \quad (26)$$

The enlarged Gaussian reference propagator is chosen as

$$\mathbb{G}_D = \begin{pmatrix} G_D & -iI \\ -iI & 0 \end{pmatrix} \quad (27)$$

with inverse

$$\mathbb{G}_D^{-1} = \begin{pmatrix} 0 & iI \\ iI & G_D \end{pmatrix} \quad (28)$$

The auxiliary cumulant interaction $\widehat{C}_D$ is defined by the exact source-transfer identity

$$e^{-i\bar{\eta}G_D\eta + \rho_D[\bar{\eta}, \eta]} = e^{-i\widehat{C}_D} \exp[-i\bar{\mathcal{J}}\mathbb{G}_D\mathcal{J}] \Big|_{\xi=\bar{\xi}=0} \quad (29)$$

Equation (29) may be regarded as the definition of $\widehat{C}_D$. It fixes all Grassmann signs and combinatorial factors unambiguously.

Expanding $\widehat{C}_D$ generates a hierarchy of auxiliary interaction vertices carrying

$$C_2^D \quad C_3^D \quad \dots \quad (30)$$

The non-Gaussianity of the Dyall reference has therefore been moved from the quadratic part of the theory into a set of many-leg auxiliary vertices.

3. Enlarged Dyson equation

The residual physical interaction acts in the physical sector, whereas the reference cumulant interaction acts in the auxiliary sector. The enlarged theory may therefore be written schematically as

$$\mathcal{V}_{\text{ext}} = V_R[\bar{\psi}_1, \psi_1] + C_D[\bar{\psi}_2, \psi_2] \quad (31)$$

Let $\mathbb{G}$ denote the exact enlarged Green's function. Since the quadratic part of the enlarged theory is Gaussian, one obtains an ordinary Dyson equation,

$$\mathbb{G}^{-1} = \mathbb{G}_D^{-1} - \bar{\Sigma} \quad (32)$$

Writing

$$\bar{\Sigma} = \begin{pmatrix} \bar{\Sigma}_{11} & \bar{\Sigma}_{12} \\ \bar{\Sigma}_{21} & \bar{\Sigma}_{22} \end{pmatrix} \quad (33)$$

the physical Green's function is simply

$$G = \mathbb{G}_{11} \quad (34)$$

The correspondence with Hall's generalized self-energy blocks is

$$\Sigma^{11} = \bar{\Sigma}_{11} \quad (35a)$$

$$\Sigma^{12} = -i\bar{\Sigma}_{12} \quad (35b)$$

$$\Sigma^{21} = -i\bar{\Sigma}_{21} \quad (35c)$$

$$\Sigma^{22} = -\bar{\Sigma}_{22} \quad (35d)$$

Block inversion of Eq. (32) yields

$$M = (1 - \Sigma^{21})^{-1} (G_D + \Sigma^{22}) (1 - \Sigma^{12})^{-1} \quad (36)$$

followed by

$$G = M + M\Sigma^{11}G \quad (37)$$

Equations (36) and (37) are precisely Hall's generalized Dyson structure and coincide with the formulation used in Ref.².

4. Physical source and enlarged vertex

Introduce a bilocal physical one-body source

$$\hat{U} = \int d(12) U(1,2) \hat{\psi}^\dagger(1) \hat{\psi}(2) \quad (38)$$

The source acts only in the physical sector, so that

$$\mathbb{U} = PUP \quad P = \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix} \quad (39)$$

We choose the source convention

$$\mathbb{G}^{-1}[U] = \mathbb{G}_D^{-1} - \mathbb{U} - \bar{\Sigma}[U] \quad (40)$$

The exact enlarged three-point vertex is defined as

$$\bar{\Gamma} = -\frac{\delta \mathbb{G}^{-1}}{\delta U} \quad (41)$$

Using

$$\delta \mathbb{G} = -\mathbb{G} (\delta \mathbb{G}^{-1}) \mathbb{G} \quad (42)$$

one obtains the exact Schwinger relation

$$\frac{\delta \mathbb{G}}{\delta U} = \mathbb{G} \bar{\Gamma} \mathbb{G} \quad (43)$$

5. Superspace Bethe–Salpeter equation

From Eq. (40),

$$\bar{\Gamma} = P + \frac{\delta \bar{\Sigma}}{\delta U} \quad (44)$$

Introduce the exact enlarged irreducible kernel

$$\bar{K}_{AB;CD} = \frac{\delta \bar{\Sigma}_{AB}}{\delta \mathbb{G}_{CD}} \quad (45)$$

The chain rule then gives

$$\bar{\Gamma} = P + \bar{K} \mathbb{G} \bar{\Gamma} \mathbb{G} \quad (46)$$

Equation (46) is the exact Bethe–Salpeter equation in the enlarged source space.

6. Exact Dyall Schwinger identity

The reference response can also be obtained directly, without invoking the enlarged formalism. Differentiating the exact Dyall Green's function with respect to the physical source gives

$$\begin{aligned} \frac{\delta G_{pq}^D(t, t')}{\delta U_{rs}(\tau)} &= G_{pr}^D(t, \tau) G_{sq}^D(\tau, t') \\ &\quad - [C_2^D]_{sp, rq}(\tau, t, \tau^+, t') \end{aligned} \quad (47)$$

In compact notation,

$$\frac{\delta G_D}{\delta U} = G_D G_D - C_2^{D, ph} \quad (48)$$

The same structure appears explicitly in the first-order expansion of Ref.².

Consequently, the enlarged reference vertex must satisfy

$$[\mathbb{G}_D \bar{\Gamma}_D \mathbb{G}_D]_{11} = G_D G_D - C_2^{D, ph} \quad (49)$$

This relation fixes the full physical projection of the reference vertex. It does *not* imply that any individual enlarged-space component, such as $\Gamma_{22,D}$, can be identified directly with C_2^D .

7. Effective physical Dyall vertex

It is convenient to define an effective physical reference vertex Γ_D through

$$G_D \Gamma_D G_D \equiv [\mathbb{G}_D \bar{\Gamma}_D \mathbb{G}_D]_{11} \quad (50)$$

Equation (49) then gives

$$G_D \Gamma_D G_D = G_D G_D - C_2^{D, ph} \quad (51)$$

Formally,

$$\Gamma_D = 1 - G_D^{-1} C_2^{D, ph} G_D^{-1} \quad (52)$$

where all integrations and particle–hole index permutations are understood.

For a correlated reference,

$$\Gamma_D \neq 1 \quad (53)$$

In the Gaussian limit,

$$C_2^D \rightarrow 0 \quad (54)$$

and therefore

$$\Gamma_D \rightarrow 1 \quad (55)$$

8. Polarization and screened residual interaction

The exact physical irreducible polarization is defined by

$$P_R = -i [\mathbb{G} \bar{\Gamma} \mathbb{G}]_{11}^{ph} \quad (56)$$

At the Dyall-reference level,

$$P_D = -i (G_D G_D - C_2^{D, ph}) \quad (57)$$

or

$$P_D = -i G_D G_D + i C_2^{D, ph} \quad (58)$$

Only the residual interaction should be screened because the active–active interaction is already contained exactly in H_D . We therefore define

$$W_R = v_R + v_R P_R W_R \quad (59)$$

At the fixed Dyall-reference level,

$$P_R \rightarrow P_D \quad (60)$$

which gives

$$W_D = v_R + v_R P_D W_D \quad (61)$$

This is the MR-RPA screened interaction employed in Ref.².

9. Exact Schwinger–Dyson hierarchy

The interaction in the enlarged theory contains two qualitatively different contributions. Define

$$\mathcal{F}_A = \frac{\delta \mathcal{V}_{\text{ext}}}{\delta \bar{\Psi}_A} \quad (62)$$

For the physical sector,

$$\mathcal{F}_1 = \frac{\delta V_R}{\delta \bar{\psi}_1} \quad (63)$$

whereas in the auxiliary sector

$$\mathcal{F}_2 = \frac{\delta C_D}{\delta \bar{\psi}_2} \quad (64)$$

Introduce the interaction insertion

$$\mathcal{X}_{AB}(1, 2) = -i \left\langle T_C \mathcal{F}_A(1) \Psi_B^\dagger(2) \right| T_C \mathcal{F}_A(1) \Psi_B^\dagger(2) \rangle \quad (65)$$

The equation of motion gives

$$\mathcal{X} = \bar{\Sigma} \mathbb{G} \quad (66)$$

The physical row therefore satisfies

$$(\bar{\Sigma} \mathbb{G})_{1B} = -i \left\langle T_C \frac{\delta V_R}{\delta \bar{\psi}_1} \Psi_B^\dagger \right| T_C \frac{\delta V_R}{\delta \bar{\psi}_1} \Psi_B^\dagger \rangle \quad (67)$$

while the auxiliary row satisfies

$$(\bar{\Sigma} \mathbb{G})_{2B} = -i \left\langle T_C \frac{\delta C_D}{\delta \bar{\psi}_2} \Psi_B^\dagger \right| T_C \frac{\delta C_D}{\delta \bar{\psi}_2} \Psi_B^\dagger \rangle \quad (68)$$

Since

$$C_D = C_D^{(2)} + C_D^{(3)} + \dots \quad (69)$$

the auxiliary row contains the complete hierarchy

$$C_2^D \quad C_3^D \quad C_4^D \quad \dots \quad (70)$$

For a term of the schematic form

$$C_D^{(n)} \propto \frac{1}{n!n!} C_n^D \bar{\psi}_2^n \psi_2^n \quad (71)$$

functional differentiation gives

$$\frac{\delta C_D^{(n)}}{\delta \bar{\psi}_2} \propto \frac{1}{(n-1)!n!} C_n^D \bar{\psi}_2^{n-1} \psi_2^n \quad (72)$$

Hence the exact multireference theory is not a closed set of five Hedin equations. Instead,

$$\text{Dyall-Hedin theory} = \text{Hedin structure} + \text{reference-cumulant hierarchy} \quad (73)$$

10. Compact form of the Dyall-Hedin hierarchy

The exact formal structure may be summarized as

$$\mathbb{G}^{-1} = \mathbb{G}_D^{-1} - \mathbb{U} - \bar{\Sigma} \quad (74a)$$

$$\frac{\delta \mathbb{G}}{\delta U} = \mathbb{G} \bar{\Gamma} \mathbb{G} \quad (74b)$$

$$\bar{\Gamma} = P + \bar{K} \mathbb{G} \bar{\Gamma} \mathbb{G} \quad (74c)$$

$$P_R = -i \left[\mathbb{G} \bar{\Gamma} \mathbb{G} \right]_{11}^{ph} \quad (74d)$$

$$W_R = v_R + v_R P_R W_R \quad (74e)$$

$$(\bar{\Sigma} \mathbb{G})_{1B} = -i \left\langle T_C \frac{\delta V_R}{\delta \bar{\psi}_1} \Psi_B^\dagger \right| T_C \frac{\delta V_R}{\delta \bar{\psi}_1} \Psi_B^\dagger \rangle \quad (74f)$$

$$(\bar{\Sigma} \mathbb{G})_{2B} = -i \left\langle T_C \frac{\delta C_D}{\delta \bar{\psi}_2} \Psi_B^\dagger \right| T_C \frac{\delta C_D}{\delta \bar{\psi}_2} \Psi_B^\dagger \rangle \quad (74g)$$

The reference vertex is constrained by

$$\left[\mathbb{G}_D \bar{\Gamma}_D \mathbb{G}_D \right]_{11} = G_D G_D - C_2^{D,ph} \quad (75)$$

B. First-order recovery of the generalized self-energy blocks

Established result. The first-order blocks derived below are the Wang–Fang–Li results. Their role here is to provide a component-level check of the exact hierarchy and to fix all signs, time orderings, and numerical prefactors.

We now verify explicitly that the Dyall-reference formalism developed in Sec. III A reproduces the four first-order generalized self-energy blocks derived by Wang, Fang, and Li².

Throughout this section,

$$G_D \equiv G_0 \quad (76)$$

denotes the exact Green’s function of the zeroth-order Hamiltonian, and

$$C_2^D \equiv G_0^c \quad C_3^D \equiv G_0^{c,(3)} \quad (77)$$

denote the connected two- and three-body zeroth-order Green’s functions.

We introduce a bookkeeping parameter λ ,

$$\hat{H}(\lambda) = \hat{H}_D + \lambda \hat{V}_R \quad (78)$$

and expand

$$G = G_D + \lambda G_1 + O(\lambda^2) \quad (79)$$

together with

$$\Sigma^{ij} = \lambda \Sigma_1^{ij} + O(\lambda^2) \quad (80)$$

1. First-order expansion of Hall's generalized Dyson equation

The generalized Dyson equation is

$$M = (1 - \Sigma^{21})^{-1}(G_D + \Sigma^{22})(1 - \Sigma^{12})^{-1} \quad (81)$$

$$G = M + M\Sigma^{11}G \quad (82)$$

To first order,

$$(1 - \Sigma^{21})^{-1} = 1 + \lambda\Sigma_1^{21} + O(\lambda^2) \quad (83)$$

and similarly for Σ^{12} . Hence

$$M = G_D + \lambda(\Sigma_1^{21}G_D + G_D\Sigma_1^{12} + \Sigma_1^{22}) + O(\lambda^2) \quad (84)$$

Substitution into the second generalized Dyson equation gives

$$G_1 = G_D\Sigma_1^{11}G_D + G_D\Sigma_1^{12} + \Sigma_1^{21}G_D + \Sigma_1^{22} \quad (85)$$

Equation (85) provides the decomposition against which the direct perturbative expansion will be matched.

2. Residual perturbation

The residual interaction is written as

$$\hat{V}_R = \hat{V}_1 + \hat{V}_2 \quad (86)$$

where

$$\hat{V}_1 = u_{pq}\hat{p}^\dagger\hat{q} \quad (87)$$

and

$$\hat{V}_2 = \frac{1}{2}v_{pr,qs}\hat{p}^\dagger\hat{q}^\dagger\hat{s}\hat{r} \quad (88)$$

Introducing the antisymmetrized interaction

$$\bar{v}_{pr,qs} = v_{pr,qs} - v_{ps,qr} \quad (89)$$

the two-body contribution may equivalently be evaluated with a 1/4 prefactor and the antisymmetrized interaction.

3. First-order Gell-Mann-Low expansion

The exact Green's function can be expanded as

$$G_{pq}(t, t') = \frac{(-i) \left\langle \mathcal{T} e^{-i \int_{t_1}^{t_2} \hat{V}_R(t_1)} \hat{p}(t) \hat{q}^\dagger(t') \right\rangle_D}{\left\langle \mathcal{T} e^{-i \int_{t_1}^{t_2} \hat{V}_R(t_1)} \right\rangle_D} \quad (90)$$

At first order,

$$G_1 = G_{1,1e} + G_{1,2e} \quad (91)$$

where the subscripts refer to the one- and two-electron pieces of $\hat{V}_R$.

4. One-body perturbation

The contribution from $\hat{V}_1$ is particularly transparent in the present formulation. Since u_{rs} couples to a one-body operator, the exact Dyall Schwinger identity derived previously immediately gives

$$[G_{1,1e}]_{pq}(t, t') = u_{rs} \int dt_1 \frac{\delta[G_D]_{pq}(t, t')}{\delta U_{rs}(t_1)} \quad (92)$$

Using

$$\frac{\delta G_D}{\delta U} = G_D G_D - C_2^D \quad (93)$$

one obtains

$$[G_{1,1e}]_{pq}(t, t') = u_{rs} \int dt_1 \left[[G_D]_{pr}(t, t_1) [G_D]_{sq}(t_1, t') - [C_2^D]_{sp,rq}(t_1, t, t_1^+, t') \right] \quad (94)$$

The first term in Eq. (94) contributes to Σ_1^{11} , whereas the second contributes to Σ_1^{22} .

5. Two-body perturbation

For the two-body perturbation, direct expansion followed by the cumulant decomposition of the three-body time-ordered Green's function gives

$$[G_{1,2e}]_{pq}(t, t') = \frac{i}{4} \bar{v}_{rt,su} \int dt_1 \left\{ [C_3^D]_{utp,qsr}(t_1^+, t_1, t, t', t_1^{++}, t_1^{+++}) - 2[G_D]_{pr}(t, t_1) [C_2^D]_{ut,sq}(t_1^+, t_1, t_1^{++}, t') - 2[G_D]_{uq}(t_1, t') [C_2^D]_{tp,sr}(t_1, t, t_1^+, t_1^{++}) - 4[G_D]_{tr}(t_1, t_1^+) [C_2^D]_{up,qs}(t_1, t, t', t_1^+) - 4[G_D]_{pr}(t, t_1) [G_D]_{us}(t_1, t_1^+) [G_D]_{tq}(t_1, t') \right\} \quad (95)$$

Here

$$t_1^{+++} > t_1^{++} > t_1^+ > t_1 \quad (96)$$

denotes the conventional infinitesimal time ordering at the interaction vertex.

The numerical coefficients in Eq. (95) are important. The two terms containing one one-particle propagator and one connected two-particle propagator carry the factor

$$\frac{i}{4}(-2) = -\frac{i}{2} \quad (97)$$

which will generate Σ_1^{12} and Σ_1^{21} .

Likewise,

$$\frac{i}{4}(-4) = -i \quad (98)$$

which generates the Hartree-Fock-like one-body contribution and the density-dependent contribution to Σ_1^{22} .

Finally, the fully connected three-body term retains the prefactor

$$\frac{i}{4} \quad (99)$$

6. Regrouping the first-order Green's function

Define the zeroth-order one-body density matrix by

$$\gamma_{pq} = \langle \hat{p}^\dagger \hat{q} | \hat{p}^\dagger \hat{q} \rangle_D = (-i)[G_D]_{qp}(t, t^+) \quad (100)$$

The effective static one-body perturbation is

$$u_{rs}^{\text{eff}} = u_{rs} + \bar{v}_{rs,tu} \gamma_{tu} \quad (101)$$

Combining Eqs. (94) and (95), and using the antisymmetry of $\bar{v}$, the first-order Green's function can be regrouped as

$$\begin{aligned} [G_1]_{pq}(t, t') = & \int dt_1 \left\{ [G_D]_{pr}(t, t_1) u_{rs}^{\text{eff}} [G_D]_{sq}(t_1, t') \right. \\ & + [G_D]_{pr}(t, t_1) \left[-\frac{i}{2} \bar{v}_{rt,su} [C_2^D]_{ut,sq}(t_1^+, t_1, t_1^{++}, t') \right] \\ & + \left[-\frac{i}{2} [C_2^D]_{pt,rs}(t, t_1, t_1^{++}, t_1^+) \bar{v}_{rt,su} \right] [G_D]_{uq}(t_1, t') \\ & - u_{rs}^{\text{eff}} [C_2^D]_{ps,qr}(t, t_1, t', t_1^+) \\ & \left. + \frac{i}{4} \bar{v}_{rt,su} [C_3^D]_{ptu,qrs}(t, t_1, t_1^+, t', t_1^{+++}, t_1^{++}) \right\} \end{aligned} \quad (102)$$

Equation (102) has exactly the four structures appearing in the first-order generalized Dyson equation, Eq. (85).

7. Identification of the four self-energy blocks

Comparison of Eqs. (85) and (102) gives the four first-order generalized self-energy blocks.

a. Physical block. The 11 block is local in time,

$$[\Sigma_1^{11}]_{rs}(t, t') = \delta(t - t') u_{rs}^{\text{eff}} \quad (103)$$

with

$$u_{rs}^{\text{eff}} = u_{rs} + \bar{v}_{rs,tu} \gamma_{tu} \quad (104)$$

b. Right mixed block. The 12 block is

$$[\Sigma_1^{12}]_{rq}(t, t') = -\frac{i}{2} \bar{v}_{rt,su} [C_2^D]_{ut,sq}(t^+, t, t^{++}, t') \quad (105)$$

c. Left mixed block. The 21 block is

$$[\Sigma_1^{21}]_{pu}(t, t') = -\frac{i}{2} [C_2^D]_{pt,rs}(t, t', t'^{++}, t'^+) \bar{v}_{rt,su} \quad (106)$$

d. Auxiliary block. Finally,

$$\begin{aligned} [\Sigma_1^{22}]_{pq}(t, t') = & \int dt_1 \left[-u_{rs}^{\text{eff}} [C_2^D]_{ps,qr}(t, t_1, t', t_1^+) \right. \\ & \left. + \frac{i}{4} \bar{v}_{rt,su} [C_3^D]_{ptu,qrs}(t, t_1, t_1^+, t', t_1^{+++}, t_1^{++}) \right] \end{aligned} \quad (107)$$

Equations (103)–(107) coincide with Eqs. (20)–(23) of Ref.².

8. Origin of the numerical prefactors

It is useful to emphasize the origin of the characteristic prefactors.

After substitution of the three-body cumulant decomposition into the antisymmetrized two-electron perturbation and combination of permutation-equivalent contractions, the terms generating the mixed blocks acquire a factor -2 , whereas the density-dependent contractions acquire a factor -4 . Combined with the overall prefactor $i/4$, these give

$$\frac{i}{4} \times (-2) = -\frac{i}{2} \quad \frac{i}{4} \times (-4) = -i \quad (108)$$

The genuinely connected three-body contribution retains the prefactor

$$+\frac{i}{4} \quad (109)$$

These factors provide a useful independent check of the enlarged-source conventions introduced in Sec. III A.

9. Specialization to the Dyall Hamiltonian

We now exploit the special structure of the Dyall reference.

The Dyall zeroth-order eigenstates factorize into inactive and active parts. The inactive sector is Gaussian, while the active sector contains the non-Gaussian correlations. Consequently, any connected zeroth-order Green's function of order two or higher vanishes if one of its orbital indices is inactive:

$$C_n^D \neq 0 \implies \text{all orbital indices of } C_n^D \text{ are active} \quad n \geq 2 \quad (110)$$

Furthermore, the residual two-electron interaction vanishes when all four indices are active,

$$v_{xy,zw}^R = 0 \quad (111)$$

10. Dyall simplification of the physical block

For the Dyall partition,

$$[\Sigma_1^{11}]_{rs} = u_{rs} + \bar{v}_{rs,pq} \gamma_{pq} \quad (112)$$

is frequency independent and Hermitian.

Partitioning the inactive orbitals into core indices $i, j, \dots$ and virtual indices $a, b, \dots$, the independent matrix elements are

$$[\Sigma_1^{11}]_{ij} = 0 \quad (113a)$$

$$[\Sigma_1^{11}]_{aj} = h_{aj} + \bar{v}_{aj,pq} \gamma_{pq} \quad (113b)$$

$$[\Sigma_1^{11}]_{xj} = h_{xj} + \bar{v}_{xj,pq} \gamma_{pq} \quad (113c)$$

$$[\Sigma_1^{11}]_{ab} = 0 \quad (113d)$$

$$[\Sigma_1^{11}]_{xb} = h_{xb} + \bar{v}_{xb,pq} \gamma_{pq} \quad (113e)$$

$$[\Sigma_1^{11}]_{xy} = 0 \quad (113f)$$

The remaining blocks follow by Hermiticity. In particular,

$$[\Sigma_1^{11}]_{xy} = 0 \quad (114)$$

because the active-active mean-field contribution already contained in the Dyall Hamiltonian exactly cancels the corresponding part of the residual one-body perturbation.

11. Vanishing of Σ_1^{22} for the Dyall reference

Equation (107) contains two terms. For the first term to be nonzero,

$$[C_2^D]_{ps,qr} \neq 0 \quad (115)$$

all four indices p, s, q, r must be active by Eq. (110). Hence

$$u_{rs}^{\text{eff}} = [\Sigma_1^{11}]_{rs} = 0 \quad (116)$$

by Eq. (114).

For the second term,

$$[C_3^D]_{ptu,qrs} \neq 0 \quad (117)$$

requires all six orbital indices to be active. In particular, r, t, s, u are active, and therefore

$$\bar{v}_{rt,su} = 0 \quad (118)$$

by Eq. (111).

In the diagrammatic formulation of Ref.², the two contributions involving the one-body and mean-field pieces cancel upon use of the Dyall relation that leads to $u_{xy}^{\text{eff}} = [\Sigma_1^{11}]_{xy} = 0$. In the regrouped expression of Eq. (107), this cancellation is already encoded in the vanishing of its first term. The remaining connected three-body term vanishes separately because the residual interaction has no all-active matrix elements, $v_{xy,zw}^R = 0$. Therefore,

$$\Sigma_1^{22} = 0 \quad (119)$$

12. Triangular structure of the mixed blocks

Consider first

$$[\Sigma_1^{12}]_{rq} = -\frac{i}{2} \bar{v}_{rt,su} [C_2^D]_{ut,sq} \quad (120)$$

A nonzero connected two-body Green's function requires

$$u, t, s, q \in A \quad (121)$$

Thus q must be active. If r were also active, all four indices of $\bar{v}_{rt,su}$ would be active, so that the residual interaction would vanish. Hence r must be inactive.

Therefore

$$\Sigma_1^{12} = \begin{pmatrix} 0 & \Sigma_1^{12,IA} \\ 0 & 0 \end{pmatrix} \quad (122)$$

Similarly,

$$[\Sigma_1^{21}]_{pu} = -\frac{i}{2} [C_2^D]_{pt,rs} \bar{v}_{rt,su} \quad (123)$$

requires

$$p, t, r, s \in A \quad (124)$$

Thus p is active. If u were also active, the residual interaction would contain four active indices and vanish. Hence u must be inactive.

Consequently,

$$\Sigma_1^{21} = \begin{pmatrix} 0 & 0 \\ \Sigma_1^{21,AI} & 0 \end{pmatrix} \quad (125)$$

Equations (122) and (125) reproduce the block structure given in Eq. (S14) of Ref.².

13. Spectral representation of the mixed blocks

The triangular structure permits particularly simple spectral representations. The expressions in this subsection are the time-ordered (Feynman) forms of Ref.²; accordingly, addition and removal poles carry opposite infinitesimal prescriptions. The conversion to a retarded representation, in which all signed poles carry the same $+i0^+$ prescription, is performed explicitly later in Sec. IV B.

Let

$$\hat{H}_D^A |\Xi_{\mu,N_a}^A\rangle = E_{\mu,N_a}^A |\Xi_{\mu,N_a}^A\rangle \quad (126)$$

and define the active charged excitation energies

$$\omega_{\mu,N_a \pm 1}^A = E_{\mu,N_a \pm 1}^A - E_{0,N_a}^A \quad (127)$$

Insertion of complete sets of $N_a + 1$ and $N_a - 1$ active-space eigenstates into Eq. (105) gives

$$\begin{aligned} [\Sigma_1^{12,IA}]_{Px}(\omega) &= \frac{1}{2} \bar{v}_{Pz,yw} \sum_{\mu} \frac{\langle \Xi_{0,N_a}^A | \hat{y}^\dagger \hat{w} \hat{z} | \Xi_{\mu,N_a+1}^A \rangle \langle \Xi_{\mu,N_a+1}^A | \hat{y}^\dagger \hat{w} \hat{z} | \Xi_{\mu,N_a+1}^A \rangle}{\omega - \omega_{\mu,N_a+1}^A + i0^+} \\ &\quad + \frac{1}{2} \bar{v}_{Pz,yw} \sum_{\mu} \frac{\langle \Xi_{0,N_a}^A | \hat{x}^\dagger | \Xi_{\mu,N_a-1}^A \rangle \langle \Xi_{\mu,N_a-1}^A | \hat{x}^\dagger | \Xi_{\mu,N_a-1}^A \rangle}{\omega + \omega_{\mu,N_a-1}^A - i0^+} \\ &\quad - \bar{v}_{Pz,yw} \langle \hat{y}^\dagger \hat{w} | \hat{y}^\dagger \hat{w} \rangle_D [G_D^A]_{zx}(\omega) \end{aligned} \quad (128)$$

Likewise,

$$\begin{aligned} [\Sigma_1^{21,AI}]_{xP}(\omega) &= -\frac{1}{2} \bar{v}_{zP,wy} \sum_{\mu} \frac{\langle \Xi_{0,N_a}^A | \hat{x} | \Xi_{\mu,N_a+1}^A \rangle \langle \Xi_{\mu,N_a+1}^A | \hat{x} | \Xi_{\mu,N_a+1}^A \rangle}{\omega - \omega_{\mu,N_a+1}^A + i0^+} \\ &\quad - \frac{1}{2} \bar{v}_{zP,wy} \sum_{\mu} \frac{\langle \Xi_{\mu,N_a-1}^A | \hat{x} | \Xi_{0,N_a}^A \rangle \langle \Xi_{0,N_a}^A | \hat{x} | \Xi_{0,N_a}^A \rangle}{\omega + \omega_{\mu,N_a-1}^A - i0^+} \\ &\quad + \bar{v}_{zP,wy} \langle \hat{w}^\dagger \hat{y} | \hat{w}^\dagger \hat{y} \rangle_D [G_D^A]_{xz}(\omega) \end{aligned} \quad (129)$$

These are Eqs. (S15) and (S16) of Ref.².

14. Interpretation of the connected subtraction

The last terms of Eqs. (128) and (129) arise from the subtraction of the disconnected part of the two-body Green's function.

For example, the attachment part of Eq. (128) contains the combination

$$\frac{1}{2} \langle \Xi_0 | \hat{y}^\dagger \hat{w} \hat{z} | \Xi_\mu^+ | \Xi_0 | \hat{y}^\dagger \hat{w} \hat{z} | \Xi_\mu^+ \rangle - \langle \hat{y}^\dagger \hat{w} | \hat{y}^\dagger \hat{w} \rangle \langle \Xi_0 | \hat{z} | \Xi_\mu^+ | \Xi_0 | \hat{z} | \Xi_\mu^+ \rangle \quad (130)$$

Thus the mixed self-energy is driven by the genuinely connected correlation between the one-particle propagation and the active-space density fluctuation. It vanishes identically for a Gaussian zeroth-order reference.

C. The existing MR-GW approximation as a Dyall-Hedin truncation

Established result. The practical MR-GW approximation in this subsection is the 2026 Wang–Fang–Li scheme. The new element is its interpretation as a channel-selective truncation of the Dyall-Hedin hierarchy derived above.

Having established the exact Dyall-Hedin hierarchy and verified its first-order expansion, we can now identify precisely the approximations that lead to the practical MR-GW method of Wang, Fang, and Li².

This connection is important for two reasons. First, it provides a functional interpretation of an approximation originally introduced through generalized diagrammatics and Hall's generalized Dyson equation. Second, it identifies unambiguously which pieces of the exact Dyall-Hedin hierarchy have been retained and which have been discarded, thereby defining a systematic path beyond the existing method.

1. One-shot Dyall-reference approximation

The exact theory contains the dressed enlarged Green's function $\mathbb{G}$, the enlarged vertex $\bar{\Gamma}$, the polarization P_R , and the residual screened interaction W_R .

The practical MR-GW approximation is one-shot. All quantities used to construct the polarization and self-energy are therefore evaluated from the fixed Dyall reference,

$$G \longrightarrow G_D \quad (131)$$

inside the self-energy construction, while the final interacting Green's function is obtained from a Dyson equation.

This is the direct multireference analogue of the G_0W_0 strategy.

2. Reference polarization

The exact Dyall-Hedin polarization is

$$P_R = -i [\mathbb{G} \bar{\Gamma} \mathbb{G}]_{11}^{ph} \quad (132)$$

At the fixed-reference level, no residual-interaction dressing of the response is retained before constructing the RPA screening. Thus

$$P_R \longrightarrow P_D = -i [\mathbb{G}_D \bar{\Gamma}_D \mathbb{G}_D]_{11}^{ph} \quad (133)$$

Using the exact Dyall Schwinger relation,

$$[\mathbb{G}_D \bar{\Gamma}_D \mathbb{G}_D]_{11} = G_D G_D - C_2^{D,ph} \quad (134)$$

one obtains

$$P_D = -i (G_D G_D - C_2^{D,ph}) \quad (135)$$

Therefore the reference polarization is *not* an independent-particle bubble.

Equivalently, introducing the effective physical Dyall vertex,

$$G_D \Gamma_D G_D = G_D G_D - C_2^{D,ph} \quad (136)$$

one may write

$$P_D = -i G_D \Gamma_D G_D \quad (137)$$

The nontrivial reference vertex Γ_D is thus retained exactly in the screening channel.

3. Equivalence with the MR-RPA irreducible polarizability

The Lehmann representation of Eq. (135) is

$$[P_D]_{pr,qs}(\omega) = \sum_{\mu>0} \frac{\langle \Phi_0 | \hat{p}^\dagger \hat{r} | \Phi_\mu \rangle \langle \Phi_0 | \hat{p}^\dagger \hat{r} | \Phi_\mu \rangle \langle \Phi_\mu | \hat{q}^\dagger \hat{s} | \Phi_0 \rangle \langle \Phi_\mu | \hat{q}^\dagger \hat{s} | \Phi_0 \rangle}{\omega - \omega_\mu + i0^+} - \sum_{\mu>0} \frac{\langle \Phi_0 | \hat{q}^\dagger \hat{s} | \Phi_\mu \rangle \langle \Phi_0 | \hat{q}^\dagger \hat{s} | \Phi_\mu \rangle \langle \Phi_\mu | \hat{p}^\dagger \hat{r} | \Phi_0 \rangle \langle \Phi_\mu | \hat{p}^\dagger \hat{r} | \Phi_0 \rangle}{\omega + \omega_\mu - i0^+} \quad (138)$$

Equation (138) is precisely the quantity denoted $\Pi_0(\omega)$ in the MR-RPA construction of Ref.². Hence,

$$\Pi_0^{\text{MR-RPA}} = P_D \quad (139)$$

This equality provides the first direct connection between the Dyall-Hedin hierarchy and the published MR-GW construction.

4. Residual screened interaction

The exact screened residual interaction satisfies

$$W_R = v_R + v_R P_R W_R \quad (140)$$

Using the fixed Dyall polarization, Eq. (133), gives

$$W_D = v_R + v_R P_D W_D \quad (141)$$

This is exactly the screened interaction employed in the practical MR-GW method: the full Coulomb interaction of conventional GW is replaced by the residual two-electron interaction, while the independent-particle RPA polarizability is replaced by the correlated Dyall response.

The active-active Coulomb interaction must not be screened again, because it is already contained nonperturbatively in H_D .

5. An important distinction from conventional GW

In conventional Hedin theory,

$$P = -iGG\Gamma \quad (142)$$

and

$$\Sigma = iGWT \quad (143)$$

so that setting

$$\Gamma \rightarrow 1 \quad (144)$$

simultaneously generates the RPA polarization and the GW self-energy.

The multireference construction is more subtle.

If one were to set

$$\Gamma_D \rightarrow 1 \quad (145)$$

in the Dyall reference response, one would obtain

$$P_D \longrightarrow -iG_D G_D \quad (146)$$

and the connected contribution

$$iC_2^{D,ph} \quad (147)$$

would be lost.

This is *not* the MR-RPA polarizability used by Ref.².

Conversely, inserting the full Γ_D explicitly into the self-energy would generate an approximation of the form

$$iG_D W_D \Gamma_D \quad (148)$$

which is also *not* the published MR-GW approximation.

The existing method therefore cannot be characterized by the single replacement

$$\Gamma \rightarrow \Gamma_D \quad (149)$$

Nor can it be characterized solely by

$$\Delta\Gamma_R = 0 \quad (150)$$

Instead, it constitutes a *channel-selective* truncation of the Dyall-Hedin hierarchy.

6. Self-energy truncation

The practical approximation retains, in the physical 11 block, the same self-energy topology as conventional one-shot GW.

From Sec. III B, the static first-order physical block is

$$[\Sigma_1^{11}]_{pq} = u_{pq} + \bar{v}_{pq,rs} \gamma_{rs}^D \quad (151)$$

The correlation contribution is obtained from the usual GW convolution, but using the interacting Dyall Green's function and the MR-RPA screened residual interaction,

$$\Sigma_{pq}^{c,MR-GW}(\omega) = \frac{i}{2\pi} \int d\omega' [G_D]_{rs}(\omega+\omega') [w_D^c]_{pr,sq}(\omega') \quad (152)$$

The resulting physical self-energy is therefore

$$\Sigma^{MR-GW} = \Sigma_1^{11} + \Sigma^{c,MR-GW} \quad (153)$$

Equation (153) is exactly the quantity defined in Eq. (S41) of Ref.².

7. Spectral form

Using the Lehmann representation of G_D and the MR-RPA spectral representation

$$[w_D^c]_{pr,qs}(\omega) = \sum_{I>0} \left[\frac{M_{pr,I} M_{qs,I}^*}{\omega - \Omega_I + i0^+} - \frac{M_{pr,I}^* M_{qs,I}}{\omega + \Omega_I - i0^+} \right] \quad (154)$$

one obtains

$$\begin{aligned} \Sigma_{pq}^{MR-GW}(\omega) = & [\Sigma_1^{11}]_{pq} \\ & + \sum_{I>0} \sum_{\mu} \frac{\langle \Phi_0 | \hat{r} | \Phi_{\mu}^{N+1} \rangle \langle \Phi_0 | \hat{r} | \Phi_{\mu}^{N+1} \rangle \langle \Phi_{\mu}^{N+1} | \hat{s}^{\dagger} | \Phi_0 \rangle \langle \Phi_{\mu}^{N+1} | \hat{s}^{\dagger} | \Phi_0 \rangle}{\omega - \Omega_I - \omega_{\mu}^{N+1} + i0^+} \\ & + \sum_{I>0} \sum_{\mu} \frac{\langle \Phi_0 | \hat{s}^{\dagger} | \Phi_{\mu}^{N-1} \rangle \langle \Phi_0 | \hat{s}^{\dagger} | \Phi_{\mu}^{N-1} \rangle \langle \Phi_{\mu}^{N-1} | \hat{r} | \Phi_0 \rangle \langle \Phi_{\mu}^{N-1} | \hat{r} | \Phi_0 \rangle}{\omega + \Omega_I + \omega_{\mu}^{N-1} - i0^+} \end{aligned} \quad (155)$$

This is the general spectral expression given in Ref.².

8. Truncation of the remaining Hall blocks

The exact generalized Dyson equation contains

$$\Sigma^{11} \quad \Sigma^{12} \quad \Sigma^{21} \quad \Sigma^{22} \quad (156)$$

The practical MR-GW approximation sets

$$\Sigma^{12} = \Sigma^{21} = 0 \quad (157)$$

For the Dyall Hamiltonian we have already established that

$$\Sigma_1^{22} = 0 \quad (158)$$

The practical approximation additionally neglects all higher-order contributions to this block,

$$\Sigma^{22} = 0 \quad (159)$$

Consequently,

$$M = G_D \quad (160)$$

Hall's generalized Dyson equation therefore collapses to

$$G = G_D + G_D \Sigma^{MR-GW} G \quad (161)$$

Equivalently,

$$G(\omega) = [G_D^{-1}(\omega) - \Sigma^{MR-GW}(\omega)]^{-1} \quad (162)$$

This is precisely the working Dyson equation of Ref.².

9. Interpretation as a channel-selective vertex truncation

The complete mapping can now be stated precisely.

In the *screening channel*, the full Dyall reference response is retained:

$$G_D \Gamma_D G_D = G_D G_D - C_2^{D,ph} \quad (163)$$

Thus

$$\Gamma_{\text{screen}} = \Gamma_D \quad (164)$$

In the *physical self-energy channel*, only the ordinary *GW* self-energy topology is retained:

$$\Sigma^{11} \longrightarrow \Sigma_1^{11} + iG_D W_D^c \quad (165)$$

where the product in the second term denotes the usual *GW* frequency convolution.

Thus no explicit reference vertex Γ_D is attached to the self-energy *GW* diagram.

Finally, in the *mixed and auxiliary channels*,

$$\Sigma^{12} = \Sigma^{21} = \Sigma^{22} = 0 \quad (166)$$

The practical method is therefore most accurately described as

$$\begin{aligned} \text{screening channel: } & \Gamma_D \text{ retained} \\ \text{11 self-energy channel: } & \text{bare } GW \text{ vertex retained} \\ \text{mixed/auxiliary channels: } & \text{discarded} \end{aligned} \quad (167)$$

This selective treatment is the precise Dyall-Hedin interpretation of the existing MR-*GW* approximation.

10. Relation to the statement $\Delta\Gamma_R = 0$

It is still useful to write formally

$$\Gamma = \Gamma_D + \Delta\Gamma_R \quad (168)$$

where $\Delta\Gamma_R$ denotes vertex corrections induced by the residual interaction.

The approximation

$$\Delta\Gamma_R = 0 \quad (169)$$

correctly expresses the fact that no additional residual-interaction vertex equation is solved when constructing P_D and W_D .

However, it is *not sufficient* to define the practical MR-*GW* self-energy, because retaining Γ_D explicitly in the self-energy would produce a $GW\Gamma_D$ -type theory.

The full definition of the published MR-*GW* approximation therefore requires both

$$\Delta\Gamma_R = 0 \quad \text{in the reference screening problem} \quad (170)$$

and the additional projected self-energy truncations

$$\Sigma^{11} \rightarrow \Sigma_1^{11} + iG_D W_D^c \quad (171)$$

together with

$$\Sigma^{12} = \Sigma^{21} = \Sigma^{22} = 0 \quad (172)$$

11. Why the construction is internally consistent

Although the vertex is treated differently in the screening and self-energy channels, this does not imply double counting.

The interaction contained completely within the active space belongs to H_D and is therefore absent from the residual interaction v_R . The strong active-space correlation consequently enters through

$$G_D \quad \text{and} \quad P_D \quad (173)$$

whereas W_D screens only the residual interaction.

The final self-energy therefore separates naturally into

nonperturbative active-space correlation+screened residual correlation (174)

This is also evident if one introduces a noninteracting propagator g_0 and defines

$$\Sigma_D = g_0^{-1} - G_D^{-1} \quad (175)$$

Then

$$G^{-1} = g_0^{-1} - (\Sigma_D + \Sigma^{\text{MR-GW}}) \quad (176)$$

which is the standard-Dyson representation discussed in Ref.².

12. MR-*GW* is not first-order exact for a correlated reference

One important consequence of the above identification should be stated explicitly.

From Sec. III B, the exact first-order Green's function is

$$G_1 = G_D \Sigma_1^{11} G_D + G_D \Sigma_1^{12} + \Sigma_1^{21} G_D + \Sigma_1^{22} \quad (177)$$

For the Dyall Hamiltonian,

$$\Sigma_1^{22} = 0 \quad (178)$$

but in general

$$\Sigma_1^{12} \neq 0 \quad \Sigma_1^{21} \neq 0 \quad (179)$$

The practical MR-*GW* approximation discards these terms. Therefore,

MR-*GW* does not reproduce the complete first-order expansion (180)

for a genuinely correlated Dyall reference.

This is not an inconsistency: MR-*GW* is a deliberately selected screened approximation rather than a finite-order perturbation theory.

However, this observation identifies immediately the lowest-cost information discarded by the existing method:

$$\Sigma^{12} \quad \text{and} \quad \Sigma^{21} \quad (181)$$

Restoring these terms in a screened and analytically consistent manner therefore provides the natural starting point for a systematic extension of MR-*GW*.

13. Objective I in compact form

The derivation establishes the following sequence:

$$\begin{aligned}
 &\text{exact Dyall-Hedin hierarchy} \\
 &\xrightarrow{\text{freeze the response at the Dyall reference}} P_D = -iG_D \Gamma_D G_D \\
 &\xrightarrow{\text{MR-RPA}} W_D = v_R + v_R P_D W_D \\
 &\xrightarrow{\text{retain only the ordinary GW topology in } \Sigma^{11}} \Sigma^{\text{MR-GW}} = \Sigma_1^{11} + iG_D W_D^c \\
 &\xrightarrow{\Sigma^{12}=\Sigma^{21}=\Sigma^{22}=0} G = G_D + G_D \Sigma^{\text{MR-GW}} G
 \end{aligned} \tag{182}$$

Thus the MR-GW method of Ref.² can be identified as a specific *channel-selective vertex and block truncation* of the Dyall-Hedin hierarchy.

This completes the first objective of the present development and, at the same time, identifies the mixed blocks Σ^{12} and Σ^{21} as the most natural targets for the first practical extension beyond the existing MR-GW approximation.

IV. OBJECTIVE II: SCREENED MIXED BLOCKS AND A HERMITIAN PSD COMPLETION

The exact and first-order analyses of Objective I identify Σ^{12} and Σ^{21} as the lowest-cost information discarded by the practical MR-GW approximation. Objective II restores these blocks in a controlled sequence: first by defining a parent-consistent static screened kernel, then by embedding the screened residues in a Hermitian pole-space representation, and finally by analyzing spectral moments and computational cost.

A. A parent-consistent static kernel for the mixed Hall sector

Proposed approximation. The static kernel introduced below is our first approximation beyond the published MR-GW scheme. It is not an exact identity: it follows from the parent MR-GW self-energy after the frozen- W and static-screening approximations.

The first extension beyond the MR-GW approximation should restore information contained in the mixed Hall blocks

$$\Sigma^{12} \quad \text{and} \quad \Sigma^{21} \tag{183}$$

while preserving the favorable analytical properties of the existing MR-GW approximation.

At first order, these blocks are generated by the antisymmetrized residual interaction,

$$\bar{v}_{pr,qs}^R = v_{pr,qs}^R - v_{ps,qr}^R \tag{184}$$

through

$$[\Sigma_1^{12}]_{rq} = -\frac{i}{2} \bar{v}_{rt,su}^R [C_2^D]_{ut,sq} \tag{185}$$

and

$$[\Sigma_1^{21}]_{pu} = -\frac{i}{2} [C_2^D]_{pt,rs} \bar{v}_{rt,su}^R \tag{186}$$

The question addressed in this section is how the bare interaction in these expressions should be screened in the cheapest approximation consistent with the parent MR-GW theory.

1. Requirements on the screened mixed kernel

We impose the following requirements.

a. Consistency with the parent MR-GW approximation. The kernel should follow from the same physical self-energy used to construct MR-GW, rather than being introduced independently.

b. Correct bare limit. When screening is removed,

$$W_D \longrightarrow v_R \tag{187}$$

the kernel must reduce to the antisymmetrized residual interaction,

$$\mathcal{I}_D^{\text{mix}} \longrightarrow \bar{v}_R \tag{188}$$

c. No additional dynamical vertex problem. The first extension should not require the solution of a new two-frequency Bethe-Salpeter equation.

d. Compatibility with a Hermitian/PSD completion. The static kernel should possess the Hermiticity properties required to construct conjugate left and right mixed couplings.

e. Correct multireference limits. The mixed contribution must vanish in the Gaussian-reference limit and in the full-active-space limit.

2. The parent MR-GW self-energy

It is useful to reorganize the physical 11 self-energy of the existing MR-GW approximation into a Hartree contribution and a GW exchange-correlation contribution.

Suppressing the one-body residual term u , which is independent of the Green's function, we may write schematically

$$\Sigma^{11,\text{MR-GW}} = \Sigma_H^R + iG_D W_D \tag{189}$$

Here

$$W_D = v_R + w_D^c \tag{190}$$

is the MR-RPA screened residual interaction.

The instantaneous bare part of $iG_D W_D$ generates the Fock exchange contribution, while $iG_D w_D^c$ generates the frequency-dependent correlation self-energy. Together with the Hartree term, Eq. (189) is equivalent to the form

$$\Sigma^{11,\text{MR-GW}} = u + \bar{v}_R \gamma_D + iG_D w_D^c \tag{191}$$

used in the explicit MR-GW expressions.

3. Functional derivative of the parent self-energy

The natural particle–hole irreducible kernel associated with the physical self-energy is

$$\mathcal{I}_D \equiv i \frac{\delta \Sigma^{11, \text{MR-GW}}}{\delta G_D} \quad (192)$$

up to the conventional placement of factors of i in the definition of the Bethe–Salpeter kernel.

Differentiating Eq. (189) generates two classes of terms:

$$\frac{\delta \Sigma_H^R}{\delta G_D} \quad (193)$$

and

$$\frac{\delta(iG_D W_D)}{\delta G_D} = iW_D + iG_D \frac{\delta W_D}{\delta G_D} \quad (194)$$

The first practical approximation is to neglect the variation of the screened interaction,

$$\frac{\delta W_D}{\delta G_D} \simeq 0 \quad (195)$$

This is the usual “frozen- W ” approximation entering the standard GW –BSE kernel.

With the conventional particle–hole index routing, one obtains

$$\begin{aligned} \mathcal{I}_D(1, 2; 3, 4) &\simeq \delta(1, 3)\delta(2, 4)v_R(1, 4) \\ &\quad - \delta(1, 4)\delta(2, 3)W_D(1, 3) \end{aligned} \quad (196)$$

Equation (196) is the direct multireference analogue of the conventional GW –BSE kernel

$$I = v - W \quad (197)$$

The two terms have different origins: the first arises from the functional derivative of the Hartree potential, while the second arises from differentiating the explicit Green’s-function line in the GW exchange-correlation self-energy.

They should therefore not be interpreted as two copies of the same screened interaction.

4. Static approximation

A fully dynamical use of Eq. (196) would introduce a second time difference into the mixed cumulant vertex.

The first-order mixed blocks contain the time-degenerate object

$$\bar{v}_R C_2^D \quad (198)$$

because the bare Coulomb interaction is instantaneous. Replacing this interaction by a fully dynamical $W_D(t, t')$ would instead require a genuine non-degenerate three-time, or equivalently two-frequency, Dyall vertex.

This would defeat the purpose of constructing the cheapest beyond-MR- GW approximation.

We therefore make the static approximation

$$W_D \longrightarrow W_D^0 \equiv W_D(\omega = 0) \quad (199)$$

The resulting static kernel is

$$\begin{aligned} \mathcal{I}_D^{\text{stat}}(1, 2; 3, 4) &= \delta(1, 3)\delta(2, 4)v_R(1, 4) \\ &\quad - \delta(1, 4)\delta(2, 3)W_D^0(1, 3) \end{aligned} \quad (200)$$

In orbital notation,

$$[\mathcal{I}_D^{\text{stat}}]_{pr,qs} = v_{pr,qs}^R - [W_D^0]_{ps,qr} \quad (201)$$

We will occasionally use the compact notation

$$\mathcal{I}_D^{\text{stat}} = v_R - (W_D^0)^x \quad (202)$$

where the superscript x denotes the crossed particle–hole index pairing.

The term “crossed” is preferable here to “direct” or “exchange”, since the latter terminology depends on the representation and index convention used for the four-point kernel.

5. Relation to the bare first-order mixed block

Using

$$W_D^0 = v_R + w_D^c(0) \quad (203)$$

Eq. (202) may be written

$$\mathcal{I}_D^{\text{stat}} = \bar{v}_R - [w_D^c(0)]^x \quad (204)$$

Thus the proposed kernel differs from the exact first-order interaction only through a static correlation correction in the crossed channel.

When screening is removed,

$$W_D^0 \longrightarrow v_R \quad (205)$$

and therefore

$$\mathcal{I}_D^{\text{stat}} \longrightarrow v_R - v_R^x = \bar{v}_R \quad (206)$$

Hence the first-order mixed self-energy derived in Sec. III B is recovered exactly.

6. Screened mixed residues

Let

$$d_{\mu+;z} = \langle \Xi_{\mu, N_a+1}^A | \hat{z}^\dagger | \Xi_{0, N_a}^A | \Xi_{\mu, N_a+1}^A | \hat{z}^\dagger | \Xi_{0, N_a}^A \rangle \quad (207)$$

and define the connected attachment tensor

$$C_{\mu; z, yw}^{D,+} = \frac{1}{2} \langle \Xi_{0, N_a}^A | \hat{y}^\dagger \hat{w} \hat{z} | \Xi_{\mu, N_a+1}^A | \Xi_{0, N_a}^A | \hat{y}^\dagger \hat{w} \hat{z} | \Xi_{\mu, N_a+1}^A \rangle - \gamma_{yw}^D d_{\mu+;z}^* \quad (208)$$

Similarly, define the removal tensor

$$C_{\mu;z,yw}^{D,-} = \frac{1}{2} \left\langle \Xi_{\mu,N_a-1}^A | \hat{y}^\dagger \hat{w} \hat{z} | \Xi_{0,N_a}^A | \Xi_{\mu,N_a-1}^A | \hat{y}^\dagger \hat{w} \hat{z} | \Xi_{0,N_a}^A \right\rangle - \gamma_{yw}^D d_{\mu;-z} \quad (209)$$

The proposed statically screened mixed residues are then

$$B_{p\mu}^{+,scr} = [\mathcal{I}_D^{\text{stat}}]_{p_{z,yw}} C_{\mu;z,yw}^{D,+} \quad (210)$$

and

$$B_{p\mu}^{-,scr} = [\mathcal{I}_D^{\text{stat}}]_{p_{z,yw}} C_{\mu;z,yw}^{D,-} \quad (211)$$

For the explicit indices appearing in the first-order mixed blocks,

$$[\mathcal{I}_D^{\text{stat}}]_{p_{z,yw}} = v_{p_{z,yw}}^R - [W_D^0]_{p_{w,yz}} \quad (212)$$

Equation (212) is therefore the practical replacement

$$\bar{v}_{p_{z,yw}}^R \longrightarrow v_{p_{z,yw}}^R - [W_D(0)]_{p_{w,yz}} \quad (213)$$

used in the first beyond-MR-GW approximation.

7. Why the fully screened antisymmetrized interaction is not the default choice

A seemingly natural alternative would be

$$\mathcal{I}_D^{\text{full-W}} = W_D^0 - (W_D^0)^x \quad (214)$$

This choice satisfies the same unscreened limit,

$$\mathcal{I}_D^{\text{full-W}} \longrightarrow \bar{v}_R \quad (215)$$

and can therefore be considered as a useful numerical control.

It is, however, not the kernel obtained from the frozen- W functional derivative of the parent MR-GW self-energy.

Indeed, the uncrossed term in Eq. (201) originates from the Hartree functional derivative and is therefore the bare residual interaction, whereas the crossed term originates from the GW exchange-correlation self-energy and contains W_D .

Replacing the first term by W_D would therefore constitute an additional approximation unrelated to the parent MR-GW functional.

For this reason,

$$\mathcal{I}_D^{\text{stat}} = v_R - (W_D^0)^x \quad (216)$$

is adopted as the default kernel.

8. Relation to positive-semidefinite vertex constructions

The same separation between a bare Coulomb interaction and a statically screened interaction appears in positive-semidefinite extensions of vertex-corrected GW .

There, the approximate four-point kernel has the structure

$$I = v - W_0^x \quad (217)$$

and the bare and statically screened interactions are treated as distinct interactions when constructing the corresponding half-amplitudes.

Our Eq. (202) is the direct Dyall-reference analogue,

$$v \rightarrow v_R \quad W_0 \rightarrow W_D(0) \quad (218)$$

This analogy is useful but should not be confused with the logical origin of the present kernel: the primary justification is the functional derivative of the parent MR-GW self-energy under the frozen- W and static approximations.

9. Positive semidefiniteness does not determine the kernel

It is important to separate two issues.

The choice

$$\mathcal{I}_D^{\text{stat}} = v_R - (W_D^0)^x \quad (219)$$

determines the *physical approximation to the mixed coupling*.

Positive semidefiniteness will instead be imposed at the next stage by embedding these couplings into a Hermitian intermediate-state representation.

Consequently, PSD alone does not select Eq. (202). Other Hermitian static kernels could also be given a PSD completion.

The present kernel is selected because it is simultaneously

- consistent with the parent MR-GW self-energy;
- obtained from the standard frozen- W functional derivative;
- compatible with the conventional static GW -BSE approximation;
- exact in the bare first-order limit;
- inexpensive;
- and compatible with the subsequent Hermitian completion.

10. Hierarchy of static mixed kernels

For numerical analysis it is useful to distinguish three related approximations.

a. *Bare mixed kernel.*

$$\mathcal{I}_D^{(0)} = \bar{v}_R \quad (220)$$

This isolates the effect of the subsequent PSD/Hermitian completion without introducing screening into the mixed block.

b. *Parent-consistent statically screened kernel.*

$$\mathcal{I}_D^{(1)} = v_R - (W_D^0)^x \quad (221)$$

This is the default approximation.

c. *Fully screened antisymmetrized kernel.*

$$\mathcal{I}_D^{(2)} = W_D^0 - (W_D^0)^x \quad (222)$$

This is retained as a useful control but is not the functional derivative of the parent MR-GW approximation.

11. Status of the approximation

The choice

$$\mathcal{I}_D^{\text{mix}} = v_R - [W_D(0)]^x \quad (223)$$

is not an exact identity of the Dyall-Hedin hierarchy.

It follows from the clearly defined sequence

$$\begin{aligned} \text{Dyall-Hedin} &\longrightarrow \text{MR-GW physical self-energy} \\ &\longrightarrow \frac{\delta \Sigma^{\text{MR-GW}}}{\delta G_D} \\ &\xrightarrow{\delta W_D / \delta G_D \rightarrow 0} v_R - W_D^x \\ &\xrightarrow{W_D(\omega) \rightarrow W_D(0)} v_R - W_D^x(0) \end{aligned} \quad (224)$$

Within the frozen-screening and static approximations, Eq. (221) is therefore the kernel consistent with the parent MR-GW theory.

Going beyond it requires restoring either

$$\frac{\delta W_D}{\delta G_D} \quad (225)$$

or the full frequency dependence of W_D , and therefore belongs to a future dynamical MR-GWT theory.

B. Common Hermitian representation and PSD completion

Proposed approximation. The finite Hermitian resolvent below defines our PSD higher-order completion. Several intermediate identities—in particular the linearization of the parent MR-GW self-energy and the first-order recovery of the mixed Hall terms—are exact algebraic statements.

We now construct a common Hermitian representation containing both the existing dynamic MR-GW self-energy and the statically screened mixed Hall blocks introduced in Sec. IV A.

The purpose of this construction is threefold:

1. to recover the existing MR-GW approximation exactly when the mixed couplings are switched off;
2. to reproduce the screened Σ^{12} and Σ^{21} contributions through first order;
3. to define a higher-order completion with a positive-semidefinite spectral function and the correct zeroth spectral moment.

The construction is most transparent for the retarded Green's function. Throughout this section,

$$z = \omega + i0^+ \quad (226)$$

1. Retarded spectral representation of the Dyall Green's function

Let n denote collectively all poles of the exact Dyall Green's function. These include

- inactive virtual poles;
- inactive core-removal poles;
- active-space $N + 1$ poles;
- active-space $N - 1$ poles.

We introduce a signed pole energy κ_n for every pole and collect these energies in the Hermitian matrix

$$K_D = \text{diag}(\kappa_1, \kappa_2, \dots) \quad (227)$$

Let t_n denote the corresponding physical transition vector. The retarded Dyall Green's function can then be written

$$G_D^R(z) = \sum_n \frac{t_n t_n^\dagger}{z - \kappa_n} \quad (228)$$

Introducing the transition matrix

$$T_D = (t_1 \ t_2 \ \dots) \quad (229)$$

and the pole-space resolvent

$$R_D(z) = (zI - K_D)^{-1} \quad (230)$$

Eq. (228) becomes

$$G_D^R(z) = T_D R_D(z) T_D^\dagger \quad (231)$$

Completeness of the exact $N + 1$ and $N - 1$ Dyall states gives the fermionic zeroth-moment sum rule

$$T_D T_D^\dagger = I \quad (232)$$

Importantly, T_D need not be square. In a correlated reference the number of Green's-function poles generally exceeds the dimension of the physical one-particle space, so that in general

$$T_D^\dagger T_D \neq I \quad (233)$$

2. Inactive and active pole states

For an inactive orbital P , there is a single zeroth-order pole with transition vector

$$t_P = e_P \quad (234)$$

where e_P is the unit vector associated with orbital P , and

$$\kappa_P = \epsilon_P \quad (235)$$

For an active charged state we introduce the composite index

$$\alpha \equiv (\mu, \sigma) \quad \sigma = \pm \quad (236)$$

with transition vector

$$d_\alpha \equiv t_{\mu\sigma} \quad (237)$$

The corresponding signed pole energies are

$$\kappa_{\mu+} = E_\mu^{N+1} - E_0^N \quad (238)$$

and

$$\kappa_{\mu-} = -(E_\mu^{N-1} - E_0^N) \quad (239)$$

Hence

$$G_D^{A,R}(z) = \sum_\alpha \frac{d_\alpha d_\alpha^\dagger}{z - \kappa_\alpha} \quad (240)$$

3. A useful linearization identity

Before introducing the mixed blocks, it is useful to establish a general identity.

Let

$$G_0(z) = TR(z)T^\dagger \quad (241)$$

with

$$TT^\dagger = I \quad (242)$$

and let $X(z)$ be an arbitrary operator in the physical one-particle space.

Then

$$T \left[R^{-1}(z) - T^\dagger X(z) T \right]^{-1} T^\dagger = \left[G_0^{-1}(z) - X(z) \right]^{-1} \quad (243)$$

To prove Eq. (243), use the Woodbury identity,

$$\begin{aligned} & \left[R^{-1} - T^\dagger X T \right]^{-1} \\ &= R + R T^\dagger X (I - T R T^\dagger X)^{-1} T R \end{aligned} \quad (244)$$

Multiplication from the left and right by T and $T^\dagger$ gives

$$\begin{aligned} T \left[R^{-1} - T^\dagger X T \right]^{-1} T^\dagger &= G_0 + G_0 X (I - G_0 X)^{-1} G_0 \\ &= (I - G_0 X)^{-1} G_0 \\ &= (G_0^{-1} - X)^{-1} \end{aligned} \quad (245)$$

Equation (243) remains valid when T is rectangular. This is crucial for an interacting reference with more poles than physical orbitals.

4. Pole representation of the MR-GW self-energy

The retarded correlation self-energy of the existing MR-GW approximation has a simple-pole representation,

$$\Sigma_{\text{MR-GW}}^{c,R}(z) = A (zI - E_{\text{GW}})^{-1} A^\dagger \quad (246)$$

where E_{GW} is a real diagonal matrix containing the signed energies of the MR-GW intermediate states.

This follows directly from the spectral expression derived in Ref.².

For example, the pole energies include

$$E_{aI} = \epsilon_a + \Omega_I \quad (247a)$$

$$E_{\mu+I} = \omega_\mu^{N+1} + \Omega_I \quad (247b)$$

$$E_{iI} = \epsilon_i - \Omega_I \quad (247c)$$

$$E_{\mu-I} = -(\omega_\mu^{N-1} + \Omega_I) \quad (247d)$$

For real orbitals and amplitudes, representative coupling amplitudes may be chosen as

$$A_{p,aI} = M_{ap,I} \quad (248a)$$

$$A_{p,\mu+I} = D_{\mu,x}^{[+1]} M_{xp,I} \quad (248b)$$

$$A_{p,iI} = M_{pi,I} \quad (248c)$$

$$A_{p,\mu-I} = D_{\mu,x}^{[-1]} M_{px,I} \quad (248d)$$

where repeated active indices are summed.

For complex quantities these definitions are understood as any factorization satisfying

$$\text{Res}_{z=E_\Lambda} \Sigma_{\text{MR-GW}}^{c,R}(z) = A_\Lambda A_\Lambda^\dagger \quad (249)$$

For a stable MR-RPA solution these residue matrices are positive semidefinite.

We separate the static contribution,

$$\Sigma_{\text{MR-GW}}^R(z) = \Sigma_{\text{stat}} + A (zI - E_{\text{GW}})^{-1} A^\dagger \quad (250)$$

where

$$\Sigma_{\text{stat}} \equiv \Sigma_1^{11} \quad (251)$$

5. Exact Hermitian linearization of the existing MR-GW approximation

Define the enlarged Hermitian matrix

$$\mathbb{H}_{\text{MR-GW}} = \begin{pmatrix} K_D + T_D^\dagger \Sigma_{\text{stat}} T_D & T_D^\dagger A \\ A^\dagger T_D & E_{\text{GW}} \end{pmatrix} \quad (252)$$

The physical transition matrix into the enlarged space is

$$\mathbb{T} = \begin{pmatrix} T_D & 0 \end{pmatrix} \quad (253)$$

Consider

$$G_{\text{lin}}^R(z) = \mathbb{T} (zI - \mathbb{H}_{\text{MR-GW}})^{-1} \mathbb{T}^\dagger \quad (254)$$

Eliminating the MR-GW bath by a Schur complement gives

$$\begin{aligned} G_{\text{lin}}^R(z) &= T_D \left[R_D^{-1}(z) - T_D^\dagger \Sigma_{\text{stat}} T_D \right. \\ &\quad \left. - T_D^\dagger A (zI - E_{\text{GW}})^{-1} A^\dagger T_D \right]^{-1} T_D^\dagger \end{aligned} \quad (255)$$

Using Eq. (243) and Eq. (250),

$$G_{\text{lin}}^R(z) = \left[(G_D^R)^{-1}(z) - \Sigma_{\text{MR-GW}}^R(z) \right]^{-1} \quad (256)$$

Thus

$$\mathbb{H}_{\text{MR-GW}} \quad (257)$$

is an exact Hermitian linearization of the published MR-GW Dyson equation.

No approximation has been introduced in passing from the pole representation of the MR-GW self-energy to Eq. (252).

6. Retarded representation of the screened mixed blocks

The statically screened mixed blocks derived in the previous section have the same active-space pole structure as the first-order blocks.

After conversion to retarded form, write

$$[\Sigma_{\text{scr}}^{12,R}]_{Px}(z) = \sum_{\alpha} \frac{B_{P\alpha}^{\text{scr}}[d_{\alpha}^{\dagger}]_x}{z - \kappa_{\alpha}} \quad (258)$$

The corresponding left block is

$$[\Sigma_{\text{scr}}^{21,R}]_{xP}(z) = \sum_{\alpha} \frac{[d_{\alpha}]_x B_{P\alpha}^{\text{scr}*}}{z - \kappa_{\alpha}} \quad (259)$$

The index α runs over *both* active addition and active removal poles.

The screened residue is

$$B_{P\alpha}^{\text{scr}} = \mathcal{I}_D^{\text{stat}} \cdot C_{P\alpha}^D \quad (260)$$

with

$$\mathcal{I}_D^{\text{stat}} = \nu_R - [W_D(0)]^x \quad (261)$$

7. Hermitian pole-space representation of the mixed blocks

Define a matrix K_B in the Dyall pole space by

$$[K_B]_{P\alpha} = B_{P\alpha}^{\text{scr}} \quad (262)$$

and

$$[K_B]_{\alpha P} = B_{P\alpha}^{\text{scr}*} \quad (263)$$

All other matrix elements are set to zero in the minimal approximation.

Therefore

$$K_B^{\dagger} = K_B \quad (264)$$

We now evaluate the first-order correction generated by K_B ,

$$\delta G_B^R(z) = T_D R_D(z) K_B R_D(z) T_D^{\dagger} \quad (265)$$

Using the explicit inactive and active pole vectors, Eq. (265) gives

$$\begin{aligned} \delta G_B^R(z) &= \sum_{P\alpha} \frac{e_P B_{P\alpha}^{\text{scr}} d_{\alpha}^{\dagger}}{(z - \kappa_P)(z - \kappa_{\alpha})} \\ &+ \sum_{P\alpha} \frac{d_{\alpha} B_{P\alpha}^{\text{scr}*} e_P^{\dagger}}{(z - \kappa_{\alpha})(z - \kappa_P)} \end{aligned} \quad (266)$$

8. Exact recovery of the mixed first-order Green's-function terms

From Eq. (258),

$$\begin{aligned} [G_D^R \Sigma_{\text{scr}}^{12,R}]_{Px} &= \frac{1}{z - \kappa_P} [\Sigma_{\text{scr}}^{12,R}]_{Px} \\ &= \sum_{\alpha} \frac{B_{P\alpha}^{\text{scr}} [d_{\alpha}^{\dagger}]_x}{(z - \kappa_P)(z - \kappa_{\alpha})} \end{aligned} \quad (267)$$

Likewise,

$$\begin{aligned} [\Sigma_{\text{scr}}^{21,R} G_D^R]_{xP} &= [\Sigma_{\text{scr}}^{21,R}]_{xP} \frac{1}{z - \kappa_P} \\ &= \sum_{\alpha} \frac{[d_{\alpha}]_x B_{P\alpha}^{\text{scr}*}}{(z - \kappa_{\alpha})(z - \kappa_P)} \end{aligned} \quad (268)$$

Comparing with Eq. (266) gives the exact identity

$$T_D R_D K_B R_D T_D^{\dagger} = G_D^R \Sigma_{\text{scr}}^{12,R} + \Sigma_{\text{scr}}^{21,R} G_D^R \quad (269)$$

Equation (269) is the central algebraic result of the present section.

A single Hermitian pole-space matrix K_B reproduces both mixed Hall blocks through first order.

9. Same-sector pole pairs

Suppose that the inactive pole P and the active pole α belong to the same physical charged sector.

For example,

$$P = a \quad \alpha = (\mu, +) \quad (270)$$

in the electron-addition sector, or

$$P = i \quad \alpha = (\mu, -) \quad (271)$$

in the electron-removal sector.

Then

$$[K_B]_{P\alpha} \quad (272)$$

has the natural interpretation of a screened coupling between Dyall states belonging to the same $N \pm 1$ sector.

In the bare limit,

$$W_D(0) \rightarrow \nu_R \quad (273)$$

the combination of this mixed residue with the corresponding off-diagonal contribution from Σ_1^{11} reconstructs the first-order residual-Hamiltonian coupling between the two charged Dyall states.

Thus the Hermitian representation has a direct charged-state interpretation for same-sector pole pairs.

10. Cross-sector pole pairs

The situation is slightly different when the two poles belong to opposite Lehmann sectors, for example

$$P = i \quad \alpha = (\mu, +) \quad (274)$$

or

$$P = a \quad \alpha = (\mu, -) \quad (275)$$

Such poles cannot be interpreted as physical states belonging to the same fixed-particle-number Hamiltonian.

Nevertheless, Eq. (269) remains valid without modification.

The reason is that the Hermitian matrix introduced here is an *auxiliary pole-space representation of the retarded Green's function*; it is not the physical $N + 1$ or $N - 1$ Hamiltonian.

All retarded poles occur with the same analytic prescription

$$\frac{1}{z - \kappa_n} \quad z = \omega + i0^+ \quad (276)$$

so a Hermitian auxiliary coupling may connect any pair of signed poles.

11. Equivalent first-order transition-amplitude representation

The cross-sector structure may also be understood by partial fractioning.

For $\kappa_P \neq \kappa_\alpha$,

$$\frac{1}{(z - \kappa_P)(z - \kappa_\alpha)} = \frac{1}{\kappa_P - \kappa_\alpha} \left[\frac{1}{z - \kappa_P} - \frac{1}{z - \kappa_\alpha} \right] \quad (277)$$

Define

$$\Delta_{P\alpha} = \kappa_P - \kappa_\alpha \quad (278)$$

For each cross-sector pair, introduce

$$S_{\alpha P} = \frac{B_{P\alpha}^{\text{scr}*}}{\Delta_{P\alpha}} \quad (279)$$

and

$$S_{P\alpha} = -\frac{B_{P\alpha}^{\text{scr}}}{\Delta_{P\alpha}} \quad (280)$$

Since the pole energies are real,

$$S^\dagger = -S \quad (281)$$

A first-order rotation of the transition matrix,

$$\delta T_D = T_D S \quad (282)$$

therefore gives

$$\delta t_P = d_\alpha \frac{B_{P\alpha}^{\text{scr}*}}{\Delta_{P\alpha}} \quad (283)$$

and

$$\delta d_\alpha = -e_P \frac{B_{P\alpha}^{\text{scr}}}{\Delta_{P\alpha}} \quad (284)$$

The corresponding residue corrections at κ_P and κ_α reproduce exactly the partial-fraction form of Eq. (266).

Thus the two representations

$$K_B \quad \text{and} \quad \delta T_D = T_D S \quad (285)$$

are equivalent through first order for cross-sector contributions.

12. Preservation of the zeroth moment in the transition-rotation representation

Because S is anti-Hermitian, the finite transformation

$$\tilde{T}_D = T_D e^S \quad (286)$$

is unitary in pole space.

Hence

$$\tilde{T}_D \tilde{T}_D^\dagger = T_D e^S e^{-S} T_D^\dagger = T_D T_D^\dagger = I \quad (287)$$

Therefore the cross-sector transition-amplitude completion preserves the zeroth spectral-moment sum rule exactly.

This provides an independent check on the sign and denominator in Eqs. (279) and (280).

13. Choice of higher-order completion

Although the Hamiltonian-coupling representation and the transition-rotation representation are equivalent through first order, their finite higher-order completions are not identical.

The present method adopts the common Hermitian pole-space coupling K_B because it

- treats same- and cross-sector pole pairs uniformly;
- combines naturally with the Hermitian MR-GW pole bath;
- preserves causality and positive spectral weight;
- and avoids a separate cross-sector implementation.

The resulting higher-order terms should therefore be regarded as a *chosen Hermitian resummation* of the screened mixed blocks, not as the unique higher-order continuation implied by the exact Dyall-Hedin hierarchy.

14. Common Hermitian matrix

We can now combine the existing MR-GW approximation with the screened mixed couplings.

Define

$$K_{\text{top}} = K_D + T_D^\dagger \Sigma_{\text{stat}} T_D + K_B \quad (288)$$

The complete auxiliary Hamiltonian is

$$\mathbb{H}_{\text{mix-MR-GW}} = \begin{pmatrix} K_{\text{top}} & T_D^\dagger A \\ A^\dagger T_D & E_{\text{GW}} \end{pmatrix} \quad (289)$$

By construction,

$$\mathbb{H}_{\text{mix-MR-GW}}^\dagger = \mathbb{H}_{\text{mix-MR-GW}} \quad (290)$$

The physical Green's function defining the proposed approximation is

$$G_{\text{mix-MR-GW}}^R(z) = \mathbb{T} (zI - \mathbb{H}_{\text{mix-MR-GW}})^{-1} \mathbb{T}^\dagger \quad (291)$$

with

$$\mathbb{T} = \begin{pmatrix} T_D & 0 \end{pmatrix} \quad (292)$$

15. Recovery of the existing MR-GW approximation

If the mixed coupling is switched off,

$$K_B = 0 \quad (293)$$

then

$$\mathbb{H}_{\text{mix-MR-GW}} \longrightarrow \mathbb{H}_{\text{MR-GW}} \quad (294)$$

Using the exact linearization derived above,

$$G_{\text{mix-MR-GW}}^R(z) \xrightarrow{K_B \rightarrow 0} \left[(G_D^R)^{-1}(z) - \Sigma_{\text{MR-GW}}^R(z) \right]^{-1} \quad (295)$$

Thus the existing MR-GW approximation is recovered exactly, not only perturbatively.

16. First-order expansion of the common Hermitian representation

To verify the perturbative limit, introduce a formal parameter λ multiplying the residual interaction.

The relevant quantities scale as

$$\Sigma_{\text{stat}} = O(\lambda) \quad (296)$$

$$K_B = O(\lambda) + \text{screening corrections} \quad (297)$$

and

$$A = O(\lambda) \quad (298)$$

so that

$$A(zI - E_{\text{GW}})^{-1} A^\dagger = O(\lambda^2) \quad (299)$$

Expansion of Eq. (291) therefore gives

$$G_{\text{mix-MR-GW}}^R = G_D^R + G_D^R \Sigma_{\text{stat}}^R G_D^R + T_D R_D K_B R_D T_D^\dagger + O(\lambda^2) \quad (300)$$

Using Eq. (269),

$$G_{\text{mix-MR-GW}}^R = G_D^R + G_D^R \Sigma_{\text{stat}}^R G_D^R + G_D^R \Sigma_{\text{scr}}^{12,R} + \Sigma_{\text{scr}}^{21,R} G_D^R + O(\lambda^2) \quad (301)$$

In the unscreened limit,

$$W_D(0) \rightarrow v_R \quad (302)$$

the screened mixed residues reduce to the exact first-order residues,

$$\Sigma_{\text{scr}}^{12} \rightarrow \Sigma_1^{12} \quad (303)$$

and

$$\Sigma_{\text{scr}}^{21} \rightarrow \Sigma_1^{21} \quad (304)$$

Since

$$\Sigma_1^{22} = 0 \quad (305)$$

for the Dyall Hamiltonian, Eq. (301) becomes

$$G_1 = G_D \Sigma_1^{11} G_D + G_D \Sigma_1^{12} + \Sigma_1^{21} G_D \quad (306)$$

which is the complete exact first-order Green's function.

Thus the Hermitian mixed-block extension restores the first-order information discarded by the practical MR-GW approximation.

17. Positive-semidefinite spectral function

Since

$$\mathbb{H}_{\text{mix-MR-GW}} = \mathbb{H}_{\text{mix-MR-GW}}^\dagger \quad (307)$$

it admits an orthonormal eigenbasis,

$$\mathbb{H}_{\text{mix-MR-GW}} U_\lambda = E_\lambda U_\lambda \quad (308)$$

Equation (291) becomes

$$G_{\text{mix-MR-GW}}^R(z) = \sum_\lambda \frac{Z_\lambda Z_\lambda^\dagger}{z - E_\lambda} \quad (309)$$

where

$$Z_\lambda = \mathbb{T} U_\lambda \quad (310)$$

Every residue matrix therefore satisfies

$$Z_\lambda Z_\lambda^\dagger \geq 0 \quad (311)$$

Consequently,

$$A(\omega) = -\frac{1}{\pi} \text{Im} G^R(\omega) \geq 0 \quad (312)$$

The proposed Hermitian completion therefore possesses a positive-semidefinite spectral function by construction.

18. Zeroth spectral moment

Completeness of the eigenvectors of $\mathbb{H}_{\text{mix-MR-GW}}$ gives

$$\sum_{\lambda} U_{\lambda} U_{\lambda}^{\dagger} = I \quad (313)$$

Hence

$$\begin{aligned} \sum_{\lambda} Z_{\lambda} Z_{\lambda}^{\dagger} &= \mathbb{T} \left(\sum_{\lambda} U_{\lambda} U_{\lambda}^{\dagger} \right) \mathbb{T}^{\dagger} \\ &= \mathbb{T} \mathbb{T}^{\dagger} \\ &= T_D T_D^{\dagger} \\ &= I \end{aligned} \quad (314)$$

Therefore

$$\int_{-\infty}^{+\infty} d\omega A(\omega) = I \quad (315)$$

Thus the Hermitian completion preserves both positive spectral weight and the exact fermionic zeroth-moment sum rule.

19. Causality

Equation (291) is the resolvent of a Hermitian matrix evaluated at

$$z = \omega + i0^+ \quad (316)$$

It therefore has no poles in the upper half of the complex frequency plane and obeys the usual retarded causality condition.

Hence the Hermitian completion simultaneously guarantees

$$\text{causality} + \text{PSD} + \text{zeroth-moment normalization} \quad (317)$$

20. Scope of the result

The following statements are exact algebraic results:

1. the matrix $\mathbb{H}_{\text{MR-GW}}$ is an exact linearization of the published MR-GW Dyson equation;

2. K_B reproduces

$$G_D \Sigma_{\text{scr}}^{12} + \Sigma_{\text{scr}}^{21} G_D \quad (318)$$

exactly through first order;

3. this statement holds for both same-sector and cross-sector pole pairs;

4. the resulting common resolvent is causal, PSD, and preserves the zeroth spectral moment.

What is *not* uniquely determined by the exact first-order theory is the higher-order resummation generated by repeated insertions of K_B .

The choice

$$(zI - \mathbb{H}_{\text{mix-MR-GW}})^{-1} \quad (319)$$

therefore defines a specific Hermitian/PSD higher-order completion of the screened mixed blocks.

This completion is selected because it preserves all known first-order constraints while avoiding the pathological incomplete resummation produced by inserting isolated first-order Hall blocks into the nonlinear generalized Dyson equation.

C. Spectral moments

Exact identity. The spectral-moment identities themselves are exact. Their evaluation with the proposed Hermitian Hamiltonian quantifies which moment constraints are preserved by the approximation.

The Hermitian representation introduced in Sec. IV B guarantees both positive spectral weight and the exact zeroth spectral moment. We now examine the first spectral moment,

$$M_1 = \int_{-\infty}^{+\infty} d\omega \omega A(\omega) \quad (320)$$

which provides a more stringent diagnostic of the approximation.

For the exact retarded Green's function, the first spectral moment is determined by the equal-time equation of motion,

$$[M_1^{\text{exact}}]_{pq} = \left\langle \left\{ [\hat{a}_p, \hat{H}] \hat{a}_q^{\dagger} \right\} \left| \left\{ [\hat{a}_p, \hat{H}] \hat{a}_q^{\dagger} \right\} \right\rangle \quad (321)$$

Consequently, the first moment probes the high-frequency static structure of the exact one-particle Green's function.

1. First moment from the high-frequency expansion

Consider a Green's function written in the Hermitian form

$$G^R(z) = \mathbb{T} (zI - \mathbb{H})^{-1} \mathbb{T}^{\dagger} \quad z = \omega + i0^+ \quad (322)$$

For large $|z|$,

$$(zI - \mathbb{H})^{-1} = \frac{I}{z} + \frac{\mathbb{H}}{z^2} + \frac{\mathbb{H}^2}{z^3} + \dots \quad (323)$$

Therefore,

$$G^R(z) = \frac{\mathbb{T} \mathbb{T}^{\dagger}}{z} + \frac{\mathbb{T} \mathbb{H} \mathbb{T}^{\dagger}}{z^2} + O(z^{-3}) \quad (324)$$

Comparison with the spectral expansion

$$G^R(z) = \frac{M_0}{z} + \frac{M_1}{z^2} + \frac{M_2}{z^3} + \dots \quad (325)$$

gives

$$M_0 = \mathbb{T} \mathbb{T}^{\dagger} \quad (326)$$

and

$$M_1 = \mathbb{T} \mathbb{H} \mathbb{T}^{\dagger} \quad (327)$$

2. First moment of the exact Dyall reference

For the Dyall reference,

$$G_D^R(z) = T_D (zI - K_D)^{-1} T_D^\dagger \quad (328)$$

Its first moment is therefore

$$M_1^D = T_D K_D T_D^\dagger \quad (329)$$

Since G_D is the exact Green's function of $\hat{H}_D$, Eq. (329) is equivalently

$$[M_1^D]_{pq} = \left\langle \left[[\hat{a}_p, \hat{H}_D] \hat{a}_q^\dagger \right] \left[[\hat{a}_p, \hat{H}_D] \hat{a}_q^\dagger \right] \right\rangle_D \quad (330)$$

3. Exact first moment for a two-body Hamiltonian

For reference, consider a Hamiltonian written as

$$\hat{H} = h_{pq} \hat{a}_p^\dagger \hat{a}_q + \frac{1}{4} \bar{v}_{pq,rs} \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r \quad (331)$$

Using Eq. (321), one obtains

$$[M_1^{\text{exact}}]_{ij} = h_{ij} + \bar{v}_{ir,js} \gamma_{rs}^{\text{exact}} \quad (332)$$

where

$$\gamma_{rs}^{\text{exact}} = \langle \hat{a}_r^\dagger \hat{a}_s | \hat{a}_r^\dagger \hat{a}_s \rangle \quad (333)$$

Thus the exact first moment depends only on the exact one-body density matrix, despite the presence of a two-body interaction.

Equivalently, if the exact self-energy has the high-frequency expansion

$$\Sigma(z) = \Sigma_\infty + \frac{\Sigma_1}{z} + \dots \quad (334)$$

then

$$\Sigma_\infty \quad (335)$$

is precisely the Hartree–Fock-like contraction of the bare interaction with the exact one-body density matrix.

4. First moment of the proposed Hermitian approximation

The common Hermitian matrix obtained in Sec. IV B is

$$\mathbb{H}_{\text{mix-MR-GW}} = \begin{pmatrix} K_{\text{top}} & T_D^\dagger A \\ A^\dagger T_D & E_{\text{GW}} \end{pmatrix} \quad (336)$$

with

$$K_{\text{top}} = K_D + T_D^\dagger \Sigma_{\text{stat}} T_D + K_B \quad (337)$$

and

$$\mathbb{T} = \begin{pmatrix} T_D & 0 \end{pmatrix} \quad (338)$$

Using Eq. (327),

$$M_1^{\text{mix-MR-GW}} = T_D K_{\text{top}} T_D^\dagger \quad (339)$$

Substitution of Eq. (337) yields

$$\begin{aligned} M_1^{\text{mix-MR-GW}} &= T_D K_D T_D^\dagger \\ &\quad + T_D T_D^\dagger \Sigma_{\text{stat}} T_D T_D^\dagger \\ &\quad + T_D K_B T_D^\dagger \end{aligned} \quad (340)$$

Using the exact zeroth-moment identity

$$T_D T_D^\dagger = I \quad (341)$$

we obtain the central result

$$M_1^{\text{mix-MR-GW}} = M_1^D + \Sigma_{\text{stat}} + T_D K_B T_D^\dagger \quad (342)$$

5. The dynamic MR-GW bath does not affect the first moment

A notable consequence of Eq. (342) is the complete absence of

$$A \quad \text{and} \quad E_{\text{GW}} \quad (343)$$

The dynamic MR-GW correlation self-energy therefore does not modify the first spectral moment.

This can also be seen directly from its high-frequency expansion,

$$\begin{aligned} \Sigma_{\text{MR-GW}}^c(z) &= A (zI - E_{\text{GW}})^{-1} A^\dagger \\ &= \frac{AA^\dagger}{z} + \frac{AE_{\text{GW}}A^\dagger}{z^2} + \mathcal{O}(z^{-3}) \end{aligned} \quad (344)$$

The dynamic correlation self-energy contains no constant term as $|z| \rightarrow \infty$.

Consequently, it first contributes to the *second* spectral moment of the Green's function rather than to the first.

6. First moment of the existing MR-GW approximation

For the practical MR-GW method,

$$K_B = 0 \quad (345)$$

Equation (342) therefore becomes

$$M_1^{\text{MR-GW}} = M_1^D + \Sigma_{\text{stat}} \quad (346)$$

The existing MR-GW approximation consequently omits any first-moment contribution associated with the mixed blocks Σ^{12} and Σ^{21} .

This provides an alternative characterization of the information discarded when these blocks are set to zero.

7. High-frequency structure of the mixed blocks

The retarded screened mixed block has the spectral representation

$$[\Sigma_{\text{scr}}^{12,R}]_{Px}(z) = \sum_{\alpha} \frac{B_{P\alpha}^{\text{scr}}[d_{\alpha}^{\dagger}]_x}{z - \kappa_{\alpha}} \quad (347)$$

At large $|z|$,

$$[\Sigma_{\text{scr}}^{12,R}]_{Px}(z) = \frac{1}{z} \sum_{\alpha} B_{P\alpha}^{\text{scr}}[d_{\alpha}^{\dagger}]_x + O(z^{-2}) \quad (348)$$

and similarly for Σ^{21} .

Since

$$G_D(z) = \frac{I}{z} + O(z^{-2}) \quad (349)$$

the contributions

$$G_D \Sigma_{\text{scr}}^{12} \quad \text{and} \quad \Sigma_{\text{scr}}^{21} G_D \quad (350)$$

begin at order

$$\frac{1}{z^2} \quad (351)$$

They therefore contribute directly to the first spectral moment.

Using the Hermitian representation derived in Sec. IV B, the sum of these contributions is exactly

$$\Delta M_1^{\text{mix}} = T_D K_B T_D^{\dagger} \quad (352)$$

8. Perturbative expansion around the Dyall Hamiltonian

Introduce again the bookkeeping parameter

$$\hat{H}(\lambda) = \hat{H}_D + \lambda \hat{V}_R \quad (353)$$

The exact first moment has the expansion

$$M_1^{\text{exact}}(\lambda) = M_1^D + \lambda M_1^{(1)} + \lambda^2 M_1^{(2)} + \dots \quad (354)$$

From Sec. III B, the exact first-order Green's function is

$$G_1 = G_D \Sigma_1^{11} G_D + G_D \Sigma_1^{12} + \Sigma_1^{21} G_D \quad (355)$$

where

$$\Sigma_1^{22} = 0 \quad (356)$$

for the Dyall Hamiltonian.

Taking the coefficient of $1/z^2$ gives

$$M_1^{(1)} = \Sigma_1^{11} + T_D K_B^{(1)} T_D^{\dagger} \quad (357)$$

where $K_B^{(1)}$ is constructed from the bare mixed residues,

$$B^{(1)} = \bar{v}_R C_D \quad (358)$$

Thus the first-order contribution from the mixed Hall blocks is required by the exact first-moment sum rule.

9. First-order exactness of the Hermitian extension

For the screened kernel,

$$\mathcal{I}_D^{\text{stat}} = v_R - [W_D(0)]^x \quad (359)$$

the screened interaction has the residual-coupling expansion

$$W_D(0) = \lambda v_R + \lambda^2 v_R P_D(0) v_R + O(\lambda^3) \quad (360)$$

Hence

$$\mathcal{I}_D^{\text{stat}} = \lambda \bar{v}_R + O(\lambda^2) \quad (361)$$

and consequently

$$K_B = \lambda K_B^{(1)} + O(\lambda^2) \quad (362)$$

Using Eq. (342),

$$M_1^{\text{mix-MR-GW}} = M_1^D + \lambda [\Sigma_1^{11} + T_D K_B^{(1)} T_D^{\dagger}] + O(\lambda^2) \quad (363)$$

Comparison with Eq. (357) gives

$$M_1^{\text{mix-MR-GW}} = M_1^{\text{exact}} + O(\lambda^2) \quad (364)$$

The proposed Hermitian screened-mixed extension is therefore *exact through first order in the residual interaction* with respect to the first spectral moment.

10. Why the existing MR-GW method is not first-order exact for the first moment

Using Eq. (346),

$$M_1^{\text{MR-GW}} = M_1^D + \lambda \Sigma_1^{11} + O(\lambda^2) \quad (365)$$

The exact result is instead

$$M_1^{\text{exact}} = M_1^D + \lambda [\Sigma_1^{11} + T_D K_B^{(1)} T_D^{\dagger}] + O(\lambda^2) \quad (366)$$

Therefore

$$M_1^{\text{MR-GW}} - M_1^{\text{exact}} = -\lambda T_D K_B^{(1)} T_D^{\dagger} + O(\lambda^2) \quad (367)$$

Unless the mixed contribution happens to vanish for a particular system or matrix element, practical MR-GW therefore violates the exact first-moment sum rule already at first residual order.

Restoring the mixed sector removes precisely this leading error.

11. Higher-order content of the screened mixed moment

The static screened kernel contains higher-order terms,

$$\mathcal{I}_D^{\text{stat}} = \lambda \bar{v}_R - \lambda^2 [v_R P_D(0) v_R]^x + \dots \quad (368)$$

Consequently,

$$T_D K_B T_D^{\dagger} \quad (369)$$

contains a selected sequence of higher-order contributions to the first moment.

These contributions should *not* be interpreted as the complete exact second- or higher-order first moment.

In particular, the exact first moment depends on the interacting one-body density matrix,

$$\gamma^{\text{exact}} = \gamma_D + \lambda \gamma^{(1)} + \lambda^2 \gamma^{(2)} + \dots \quad (370)$$

whereas the one-shot MR-GW construction does not determine all such density-response terms self-consistently.

Therefore,

the screened Hermitian approximation is first-order exact in M_1 but not generally exact beyond first order. (371)

12. The role of the dynamic MR-GW bath in higher moments

Although the MR-GW bath does not enter M_1 , it enters the next moment.

From

$$M_2 = \text{TH}^2 \text{T}^\dagger \quad (372)$$

one obtains

$$M_2 = T_D K_{\text{top}}^2 T_D^\dagger + A A^\dagger \quad (373)$$

Thus the dynamic MR-GW amplitudes first appear through

$$A A^\dagger \quad (374)$$

in the second spectral moment.

This provides a useful interpretation of the division of roles:

M_0 : normalization

M_1 : static and mixed-reference couplings

M_2 and higher : dynamic MR-GW screening enters explicitly (375)

13. First-moment error as a practical diagnostic

Equation (321) suggests a useful numerical diagnostic.

Given an approximate one-body density matrix γ obtained from the resulting Green's function, one may independently construct

$$M_1^{\text{comm}}[\gamma] \quad (376)$$

from the equal-time commutator with the physical Hamiltonian.

This can be compared with the spectral moment

$$M_1^{\text{spec}} = \int d\omega \omega A(\omega) \quad (377)$$

The difference

$$\Delta_1 = M_1^{\text{spec}} - M_1^{\text{comm}}[\gamma] \quad (378)$$

provides a compact measure of the internal moment consistency of a given truncation.

For the exact Green's function,

$$\Delta_1 = 0 \quad (379)$$

This comparison may be particularly useful when assessing the three candidate mixed kernels

$$\bar{v}_R \quad v_R - W_D^x(0) \quad W_D(0) - W_D^x(0) \quad (380)$$

D. Computational formulation and scaling

but not generally exact beyond first order.

Proposed approximation. This subsection turns the formal construction into a practical algorithm. Costs quoted for the parent spectral MR-GW implementation are those of Ref.²; the additional mixed-block costs are estimates for the implementation proposed here.

We finally discuss the practical implementation of the screened-mixed MR-GW approximation developed in the preceding sections.

The objective is not to design a fully optimized low-scaling implementation at this stage, but rather to establish that the new mixed-block contribution can be added to the existing MR-GW algorithm at moderate additional cost.

We follow the notation of Ref.². Let

$$N_{\text{orb}} = N_c + N_a + N_v \quad (381)$$

where N_c , N_a , and N_v denote the number of core, active, and virtual spin orbitals, respectively.

We further define

$$N_I = N_c + N_v \quad (382)$$

as the number of inactive orbitals.

Let

$$D_0 \quad D_+ \quad D_- \quad (383)$$

be the determinant dimensions of the neutral, electron-attached, and electron-removed active sectors, and define

$$D = \max(D_0, D_+, D_-) \quad (384)$$

The dimension of each MR-RPA A or B block is denoted by K , with the bound

$$K \leq N_c N_v + D(N_c + N_v + 1) \quad (385)$$

Finally, let L denote the number of charged channels retained in the Dyall one-particle Green's function. For a full spectral representation,

$$L \leq N_c + N_v + D_+ + D_- \leq N_c + N_v + 2D \quad (386)$$

The second inequality is the bound used in Ref.²; the first allows for symmetry-forbidden or otherwise omitted charged channels.

1. Quantities already required by MR-GW

The existing spectral MR-GW implementation already provides the following quantities:

$$E_{0,N_a}^A \quad E_{\mu,N_a}^A \quad E_{\mu,N_a+1}^A \quad E_{\mu,N_a-1}^A \quad (387)$$

together with the transition amplitudes

$$D_{\mu,x}^{[+1]} = \langle \Xi_{\mu,N_a+1}^A | \hat{x}^\dagger | \Xi_{0,N_a}^A | \Xi_{\mu,N_a+1}^A | \hat{x}^\dagger | \Xi_{0,N_a}^A \rangle \quad (388a)$$

$$D_{\mu,x}^{[-1]} = \langle \Xi_{\mu,N_a-1}^A | \hat{x} | \Xi_{0,N_a}^A | \Xi_{\mu,N_a-1}^A | \hat{x} | \Xi_{0,N_a}^A \rangle \quad (388b)$$

$$D_{\mu,xy}^{[0]} = \langle \Xi_{\mu,N_a}^A | \hat{x}^\dagger \hat{y} | \Xi_{0,N_a}^A | \Xi_{\mu,N_a}^A | \hat{x}^\dagger \hat{y} | \Xi_{0,N_a}^A \rangle \quad (388c)$$

These quantities are used to construct the MR-RPA matrices, whose diagonalization yields

$$\Omega_I \quad X_{\mu I} \quad Y_{\mu I} \quad (389)$$

The screened-interaction amplitudes are

$$M_{pr,I} = v_{pr,qs}^R \left[\langle \Phi_0 | \hat{q}^\dagger \hat{s} | \Phi_\mu | \Phi_0 | \hat{q}^\dagger \hat{s} | \Phi_\mu \rangle X_{\mu I} + \langle \Phi_\mu | \hat{q}^\dagger \hat{s} | \Phi_0 | \Phi_\mu | \hat{q}^\dagger \hat{s} | \Phi_0 \rangle Y_{\mu I} \right] \quad (390)$$

No modification of these parts of the MR-GW calculation is required by the present extension.

2. Cost of the parent spectral MR-GW calculation

For reference, the dense spectral implementation of Ref.² has the following principal costs:

Operation	Leading cost
Four-index AO-MO transformation	$O(N_{\text{orb}}^5)$
Active-sector diagonalizations	$O(D^3)$
Active transition densities	$O(D^2 N_a^2)$
Dense MR-RPA diagonalization	$O(K^3)$
Construction of all $M_{pr,I}$	$O(N_{\text{orb}}^2 K^2)$
MR-GW self-energy on a frequency grid	$O(N_\omega N_\Sigma K L)$
Physical Dyson inversion	$O(N_\omega N_{\text{orb}}^3)$

Without an active space,

$$K = O(N_{\text{orb}}^2) \quad (391)$$

so the dense spectral implementation has the conventional

$$O(N_{\text{orb}}^6) \quad (392)$$

scaling of a full-pole dense $G_0 W_0$ implementation.

At fixed active space, the same asymptotic orbital-space scaling is retained in MR-GW, with additional determinant-space costs associated with the active reference.

3. Static screened interaction required by the new method

The new mixed kernel requires only the zero-frequency screened interaction,

$$W_D^0 = W_D(0) \quad (393)$$

Using the MR-RPA spectral representation,

$$[w_D^c]_{pr,qs}(\omega) = \sum_{I>0} \left[\frac{M_{pr,I} M_{qs,I}^*}{\omega - \Omega_I + i0^+} - \frac{M_{pr,I}^* M_{qs,I}}{\omega + \Omega_I - i0^+} \right] \quad (394)$$

one obtains at zero frequency

$$[w_D^c(0)]_{pr,qs} = - \sum_{I>0} \frac{M_{pr,I} M_{qs,I}^* + M_{pr,I}^* M_{qs,I}}{\Omega_I} \quad (395)$$

For real quantities,

$$[w_D^c(0)]_{pr,qs} = -2 \sum_{I>0} \frac{M_{pr,I} M_{qs,I}}{\Omega_I} \quad (396)$$

Only the mixed slice

$$[W_D^0]_{Pw,yz} \quad (397)$$

is required.

If this slice is formed explicitly from the already available $M_{pr,I}$, its cost is

$$O(K N_I N_a^3) \quad (398)$$

with storage

$$O(N_I N_a^3) \quad (399)$$

At fixed active space this is only cubic in the external orbital dimension and is therefore subleading relative to the dense MR-RPA step.

4. Avoiding the explicit three-operator transition tensor

A direct implementation of the screened mixed residues might first construct

$$C_{\mu;z,yw}^{D,+} \quad (400)$$

and

$$C_{\mu;z,yw}^{D,-} \quad (401)$$

Their storage would scale as

$$O(D N_a^3) \quad (402)$$

and transformation to all dense charged eigenstates would require at most

$$O(D^2 N_a^3) \quad (403)$$

work.

This explicit tensor is unnecessary.

Define instead, for every inactive orbital P , the active-space operator

$$\hat{Q}_P = \sum_{zyw} [\mathcal{I}_D^{\text{stat}}]_{Pz,yw} \left[\frac{1}{2} \hat{y}^\dagger \hat{w} \hat{z} - \gamma_{yw}^D \hat{z} \right] \quad (404)$$

The screened mixed residues can then be evaluated directly as

$$B_{P\mu}^{+,scr} = \langle \Xi_{0,N_a}^A | \hat{Q}_P | \Xi_{\mu,N_a+1}^A | \Xi_{0,N_a}^A | \hat{Q}_P | \Xi_{\mu,N_a+1}^A \rangle \quad (405a)$$

$$B_{P\mu}^{-,scr} = \langle \Xi_{\mu,N_a-1}^A | \hat{Q}_P | \Xi_{0,N_a}^A | \Xi_{\mu,N_a-1}^A | \hat{Q}_P | \Xi_{0,N_a}^A \rangle \quad (405b)$$

Equivalently, introduce the charged source vectors

$$|\chi_P^+\rangle = \hat{Q}_P^\dagger |\Xi_{0,N_a}^A\rangle \quad (406a)$$

$$|\chi_P^-\rangle = \hat{Q}_P |\Xi_{0,N_a}^A\rangle \quad (406b)$$

Then

$$B_{P\mu}^{+,scr} = \langle \chi_P^+ | \Xi_{\mu,N_a+1}^A | \chi_P^+ | \Xi_{\mu,N_a+1}^A \rangle \quad (407a)$$

$$B_{P\mu}^{-,scr} = \langle \Xi_{\mu,N_a-1}^A | \chi_P^- | \Xi_{\mu,N_a-1}^A | \chi_P^- \rangle \quad (407b)$$

This formulation avoids storing any object with indices (μ, z, y, w) .

5. Cost of the contracted-source construction

A general three-active-index operator connects a determinant to at most $O(N_a^3)$ determinants.

Constructing one source vector therefore requires at most

$$O(DN_a^3) \quad (408)$$

operations.

For all inactive orbitals,

$$O(N_I DN_a^3) \quad (409)$$

work is sufficient to form the source vectors.

Projection of each source vector onto all charged eigenstates requires at most

$$O(D^2) \quad (410)$$

work per inactive orbital.

Hence a conservative upper bound for direct construction of all mixed residues is

$$O[N_I (DN_a^3 + D^2)] \quad (411)$$

The resulting residues require only

$$O[N_I (D_+ + D_-)] \quad (412)$$

storage.

If only a subset of charged states is required, D in the projection step can be replaced by the number of retained charged eigenstates.

6. Factorized evaluation of the screened contribution

The correlation part of the static screened interaction has a low-rank MR-RPA representation.

For the crossed slice entering the mixed kernel,

$$[w_D^c(0)]_{Pw,yz} = -2 \sum_{I>0} \frac{M_{Pw,I} M_{yz,I}}{\Omega_I} \quad (413)$$

for real quantities.

Consequently, it is not necessary to construct $[W_D^0]_{Pw,yz}$ explicitly.

The contraction with the connected active tensor may instead be factorized by first defining

$$F_{\mu w I}^\sigma = \sum_{yz} M_{yz,I} C_{\mu; z, yw}^{D, \sigma} \quad (414)$$

The correlation part of the residue then takes the form

$$B_{P\mu}^{\sigma,c} = 2 \sum_{Iw} \frac{M_{Pw,I} F_{\mu w I}^\sigma}{\Omega_I} \quad (415)$$

up to the sign associated with the kernel convention of Sec. IV A.

A direct factorized implementation therefore has the schematic cost

$$O(KDN_a^3 + N_I KDN_a) \quad (416)$$

plus the charged-state projection cost when the transition tensors are generated from source vectors.

The MR-RPA mode index I can be blocked, so the intermediate $F_{\mu w I}^\sigma$ need never be stored for all modes simultaneously.

7. Construction and storage of K_B

Once the screened residues have been obtained, the Hermitian mixed matrix is defined by

$$[K_B]_{P\alpha} = B_{P\alpha}^{scr} \quad [K_B]_{\alpha P} = B_{P\alpha}^{scr*} \quad (417)$$

The only nonzero elements connect inactive poles to active charged poles.

Therefore,

$$N_{nz}(K_B) = O[N_I (D_+ + D_-)] \quad (418)$$

and the matrix-vector product

$$K_B x \quad (419)$$

requires the same order of work.

Thus K_B is a very sparse object even though the complete auxiliary Hamiltonian is large.

8. Size of the MR-GW pole bath

The spectral MR-GW self-energy contains, for every positive MR-RPA mode I , the charged channels

$$a \quad i \quad (\mu, +) \quad (\mu, -) \quad (420)$$

The number of self-energy poles is therefore bounded by

$$N_{\text{bath}} \lesssim KL \quad (421)$$

The formal Hermitian matrix of Sec. IV B consequently has dimension

$$N_H = L + N_{\text{bath}} = O(KL) \quad (422)$$

A dense construction or dense diagonalization of this matrix would be prohibitively expensive and is neither necessary nor recommended.

9. Matrix-free application of the common Hermitian matrix

The common Hermitian matrix is

$$\mathbb{H} = \begin{pmatrix} K_D + T_D^\dagger \Sigma_{\text{stat}} T_D + K_B & T_D^\dagger A \\ A^\dagger T_D & E_{\text{GW}} \end{pmatrix} \quad (423)$$

For a trial vector

$$x = \begin{pmatrix} x_D \\ x_{\text{bath}} \end{pmatrix} \quad (424)$$

its action may be evaluated as

$$y_D = K_D x_D + K_B x_D + T_D^\dagger \Sigma_{\text{stat}} (T_D x_D) + T_D^\dagger (A x_{\text{bath}}) \quad (425a)$$

$$y_{\text{bath}} = E_{\text{GW}} x_{\text{bath}} + A^\dagger (T_D x_D) \quad (425b)$$

Thus neither

$$T_D^\dagger A \quad (426)$$

nor the full matrix $\mathbb{H}$ needs to be stored.

If the self-energy amplitudes A are treated as an explicit rectangular matrix, a matrix-vector product requires approximately

$$O(N_{\text{orb}} KL) \quad (427)$$

work, in addition to the much cheaper sparse K_B multiplication.

At fixed active space,

$$K = O(N_{\text{orb}}^2) \quad L = O(N_{\text{orb}}) \quad (428)$$

so

$$C_{\text{matvec}} = O(N_{\text{orb}}^4) \quad (429)$$

This remains below the $O(N_{\text{orb}}^6)$ cost of the dense MR-RPA spectral implementation.

10. Factorized application of the MR-GW bath

It is not necessary to form A explicitly either.

The entries of A factor into the already available screened interaction amplitudes $M_{pr,I}$ and the Dyall addition/removal transition densities.

For example, the active electron-addition amplitudes contain

$$D_{\mu,x}^{[+1]} M_{xp,I} \quad (430)$$

while the removal amplitudes contain

$$D_{\mu,x}^{[-1]} M_{px,I} \quad (431)$$

Therefore the actions

$$A x_{\text{bath}} \quad (432)$$

and

$$A^\dagger y \quad (433)$$

can be implemented directly from the same precontracted objects used to evaluate the MR-GW self-energy.

The Hermitian formulation therefore does not require storing an additional object of dimension

$$N_{\text{orb}} \times KL \quad (434)$$

11. Two practical solution strategies

The common Hermitian representation can be exploited in two complementary ways.

For selected charged excitations, one may apply Lanczos, Davidson, or another iterative Hermitian eigensolver directly to $\mathbb{H}$ using only the matrix-vector products described above.

For a frequency-dependent spectral function, it is preferable to eliminate the MR-GW bath analytically. The Schur complement gives

$$R_{\text{top}}^{-1}(z) = zI - K_D - T_D^\dagger \Sigma_{\text{stat}} T_D - K_B - T_D^\dagger A (zI - E_{\text{GW}})^{-1} A^\dagger T_D \quad (435)$$

The physical Green's function is then

$$G^R(z) = T_D R_{\text{top}}(z) T_D^\dagger \quad (436)$$

Equation (435) involves only an L -dimensional top space. The large bath enters exclusively through the action of the factorized dynamic self-energy.

This is likely the preferred route for computing spectral functions on a frequency grid.

12. Scaling of the new ingredients

The principal additional operations introduced relative to the parent MR-GW calculation are summarized below:

New operation	Work	Storage	This should be compared with the existing active-sector storage and matrix-free	
Explicit mixed $W_D(0)$ slice	$O(KN_I N_a^3)$	$O(N_I N_a^3)$		
Explicit charged three-operator tensors	$O(D^2 N_a^3)$	$O(DN_a^3)$	$O(D^3)$	(446)
Contracted charged source vectors	$O[N_I(DN_a^3 + D^2)]$	$O(N_I D)$	and MR-RPA determinant dependence through blockable	
Factorized screened contraction	$O(KDN_a^3 + N_I KDN_a)$	$O(N_I D)$		
Construction/storage of K_B	$O(N_I D)$	$O(N_I D)$	$K \leq N_c N_v + D(N_c + N_v + 1)$	(447)
Hermitian matrix-vector product	$O(N_{\text{orb}} KL)$	matrix-free		

The explicit three-operator tensor and the explicit $W_D(0)$ slice are alternative implementation routes rather than mandatory simultaneous costs.

The recommended implementation uses contracted source vectors and factorized MR-RPA amplitudes wherever possible.

13. Asymptotic orbital scaling at fixed active space

For fixed N_a and fixed determinant dimensions,

$$D = O(1) \quad (437)$$

while

$$N_I = O(N_{\text{orb}}) \quad (438)$$

$$L = O(N_{\text{orb}}) \quad (439)$$

and

$$K = O(N_{\text{orb}}^2) \quad (440)$$

The new static mixed-screening operations then scale no worse than approximately

$$O(N_{\text{orb}}^3) \quad (441)$$

while a matrix-free application of the final Hermitian problem scales as

$$O(N_{\text{orb}}^4) \quad (442)$$

Both are subleading relative to the

$$O(N_{\text{orb}}^6) \quad (443)$$

dense MR-RPA diagonalization already present in the spectral implementation.

Consequently,

the screened-mixed PSD extension does not increase the leading orbital-space scaling of dense spectral MR-GW (444)

14. Determinant-space scaling

The principal genuinely new determinant-space quantity is the three-active-operator source vector.

Using the contracted-source formulation, its conservative cost is

$$O[N_I(DN_a^3 + D^2)] \quad (445)$$

For the small and moderate complete active spaces targeted by the initial implementation, this additional cost should therefore remain manageable.

For substantially larger active spaces, the same source-vector formulation is naturally compatible with iterative CI or DMRG representations because it only requires the action of $\hat{Q}_P$ on the reference wave function and overlaps with charged response states.

15. Relation to a future frequency-domain implementation

The spectral implementation is useful for developing and validating the present theory, but it should not be regarded as the ultimate algorithm.

Ref.² already describes an alternative frequency-domain formulation in which the active neutral and charged responses are obtained from sparse linear equations rather than full diagonalization.

In that formulation the active response scales schematically as

$$O(N_{\omega} N_{\text{it}} D N_a^5 N_{\text{orb}}) \quad (448)$$

while modern RI-based orbital-space contractions can reduce selected MR-GW quantities to the same asymptotic scaling as modern single-reference GW.

The mixed source vectors introduced here are particularly compatible with such an approach because they are already expressed as operator actions,

$$\hat{Q}_P |\Xi_0\rangle \quad (449)$$

rather than requiring complete charged-state eigenvectors.

Thus a future low-scaling formulation can evaluate the screened mixed sector directly through charged-sector resolvents without constructing all $B_{P\mu}^\sigma$ explicitly.

16. Recommended initial implementation

For a first proof-of-principle implementation, the following workflow is sufficient:

1. Perform CASCI or CASSCF and construct the Dyall Hamiltonian.
2. Diagonalize the neutral and $N_a \pm 1$ active sectors, as in the existing MR-GW implementation.
3. Construct the standard transition densities $D^{[0]}$, $D^{[+1]}$, and $D^{[-1]}$.

4. Solve the dense MR-RPA problem and obtain Ω_I , X_I , Y_I , and $M_{pr,I}$.
5. Construct only the static mixed kernel required for

$$\mathcal{I}_D^{\text{stat}} = v_R - W_D^x(0) \quad (450)$$

6. Build the contracted source operators $\hat{Q}_P$ and obtain the screened residues $B_{P\mu}^{\pm, \text{scr}}$ without forming the full three-operator transition tensor.
7. Assemble the sparse Hermitian matrix K_B .
8. Reuse the existing MR-GW pole amplitudes to apply the dynamic bath matrix-free.
9. Compute either selected poles with an iterative Hermitian eigensolver or the spectral function through the top-space Schur complement of Eq. (435).
10. Verify numerically the PSD property, zeroth spectral moment, first spectral moment, and all limiting cases derived in the preceding sections.

17. Quantities that should not be formed explicitly

Although the formal derivation uses several large matrices, an implementation should avoid explicitly constructing

$$\mathbb{H}_{\text{mix-MR-GW}} \quad (451)$$

$$T_D^\dagger A \quad (452)$$

the complete $N_{\text{orb}} \times KL$ self-energy amplitude matrix, and the complete three-operator transition tensor

$$C_{\mu; z, yw}^{D, \sigma} \quad (453)$$

All of these objects admit factorized or matrix-free actions.

The only genuinely new object that should normally be stored is the sparse inactive-active coupling matrix

$$K_B \quad (454)$$

V. CONSISTENCY LIMITS AND HIERARCHY OF APPROXIMATIONS

The formal hierarchy and the proposed approximation are constrained by a set of limiting cases that should be regarded as non-negotiable implementation tests.

A. No residual interaction and full active space

Exact identity. If

$$\hat{V}_R = 0 \quad (455)$$

then all residual self-energy and screening corrections vanish and

$$G = G_D \quad (456)$$

The Dyall cumulants may remain nonzero and continue to enter response functions, but they do not generate an additional physical two-point self-energy in the absence of a residual interaction.

If the active space spans the complete one-particle space, then

$$\hat{H}_D = \hat{H} \quad \hat{V}_R = 0 \quad (457)$$

so that

$$G_D = G_{\text{exact}} \quad (458)$$

B. Gaussian or empty-active-space limit

Exact identity. When the Dyall reference becomes Gaussian,

$$C_{n \geq 2}^D = 0 \quad (459)$$

The auxiliary cumulant hierarchy disappears,

$$\Sigma^{12} = \Sigma^{21} = \Sigma^{22} = 0 \quad (460)$$

while

$$\Gamma_D \rightarrow 1 \quad P_D \rightarrow -iG_D G_D \quad (461)$$

The generalized Dyson equation therefore reduces to the ordinary Dyson equation and the formal hierarchy collapses to conventional Hedin theory. For an empty active space, the practical MR-GW construction correspondingly reduces to the single-reference $G_0 W_0$ limit associated with the chosen one-particle reference.

C. Recovery of the published MR-GW scheme

Established result. In the proposed Hermitian representation, setting

$$K_B = 0 \quad (462)$$

removes the mixed sector exactly and recovers the Wang-Fang-Li MR-GW Dyson equation and its pole-space linearization.

D. Bare mixed-block limit

Exact identity. At leading residual order,

$$W_D(0) \rightarrow v_R \quad (463)$$

implies

$$\mathcal{I}_D^{\text{stat}} \rightarrow \bar{v}_R \quad (464)$$

The screened residues therefore reduce to the exact first-order mixed self-energy blocks derived in Sec. III B. Since $\Sigma_1^{22} = 0$ for the Dyall partition, the Hermitian mixed extension recovers the complete first-order Green's function in this limit.

E. PSD, causality, and spectral moments

Proposed approximation. For the finite Hermitian completion of Sec. IV B, Hermiticity guarantees a retarded causal resolvent with residue matrices of the form

$$Z_\lambda Z_\lambda^\dagger \geq 0 \quad (465)$$

Consequently,

$$A(\omega) \geq 0 \quad \int d\omega A(\omega) = I \quad (466)$$

The first moment satisfies

$$M_1 = M_1^D + \Sigma_1^{11} + T_D K_B T_D^\dagger \quad (467)$$

and is exact through first order in the residual interaction. Higher-order terms generated by static screening and repeated insertions of K_B define the chosen approximation and are not claimed to constitute the complete higher-order Dyall-Hedin expansion.

F. Hierarchy

The resulting sequence of approximations can be summarized as

$$\begin{array}{c} \text{exact Dyall-Hedin hierarchy} \\ \Downarrow \\ \text{published MR-GW} \quad (\Sigma^{12} = \Sigma^{21} = \Sigma^{22} = 0) \\ \Downarrow \\ \text{static screened mixed extension} \quad [I_D^{\text{stat}} = v_R - W_D^x(0)] \\ \Downarrow \\ \text{Hermitian/PSD completion} \\ \Downarrow \\ \text{future dynamical MR-GWT theory} \end{array} \quad (468)$$

VI. CONCLUSIONS AND OUTLOOK

These notes develop two connected pieces of a multireference Green's-function program based on a CASSCF wave function and the Dyall Hamiltonian.

For Objective I, the exact non-Gaussian Dyall reference is embedded in an enlarged Gaussian source space in which an ordinary matrix Dyson equation and a superspace Bethe–Salpeter equation can be written. The non-Gaussian character of the reference reappears as a hierarchy of auxiliary cumulant

vertices. The physical projection reproduces Hall's generalized Dyson equation, the exact source derivative gives

$$\frac{\delta G_D}{\delta U} = G_D G_D - C_2^{D,ph} \quad (469)$$

and the resulting first-order expansion reproduces the four generalized self-energy blocks of Wang, Fang, and Li. Their practical MR-GW method is then identified as a channel-selective truncation: the correlated Dyall response is retained in the screening channel, the ordinary GW topology is used in the physical 11 self-energy, and the mixed and auxiliary Hall blocks are discarded.

For Objective II, the leading discarded mixed blocks are restored through the static kernel

$$I_D^{\text{stat}} = v_R - W_D^x(0) \quad (470)$$

obtained from the parent MR-GW self-energy under frozen- W and static approximations. The resulting mixed residues are embedded together with the dynamic MR-GW pole bath in a common Hermitian auxiliary Hamiltonian. This construction reproduces the mixed Green-function terms through first order, is causal and positive semidefinite, preserves the zeroth spectral moment, and restores the complete first-order contribution to the first moment. Importantly, the additional operations do not increase the leading dense orbital-space scaling of the parent spectral implementation at fixed active space.

Several extensions are natural. The first is numerical: implement the bare, parent-consistent static, and fully screened static mixed kernels and compare their spectral properties, quasiparticle energies, and moment errors on systems where the neglected Hall blocks are appreciable. The second is formal: replace the static kernel by the full two-frequency object

$$W_D(\nu) \Gamma_D(\omega, \nu) \quad (471)$$

and ultimately restore residual vertex corrections $\delta W_D / \delta G_D$ and the associated mixed Bethe–Salpeter equations. A third direction is to clarify how the Hermitian completion reorganizes higher-order contributions involving Σ^{12} , Σ^{21} , and Σ^{22} without identifying that completion with any single Hall block.

The practical philosophy is therefore deliberately incremental: retain the exact CASSCF/Dyall reference, reuse the MR-RPA screening already available in MR-GW, restore the cheapest missing mixed-sector information first, and increase the vertex content only when required by accuracy.

¹C. Brouder, in *Quantum Field Theory: Competitive Models*, edited by B. Fauser, J. Tolksdorf, and E. Zeidler (Birkhäuser, Basel, 2009) pp. 163–175, arXiv:0710.5652.

²Y. Wang, W.-H. Fang, and Z. Li, “Multi-reference gw approximation for strongly correlated molecules,” (2026), arXiv:2604.16013, arXiv:2604.16013 [physics.chem-ph].